

# Diagnostic Inflation Expectations in China: Salience, Overreaction, and the Limits of Rational Updating\*

Jingle Fu<sup>†</sup>

March 2026

## Abstract

Do Chinese households overreact to salient inflation signals? Using seven waves of the China Family Panel Studies ( $N \approx 13,700$ ), I estimate a revision-error coefficient of  $\hat{\beta} \approx -0.55$  ( $p < 0.001$ ) with province and wave fixed effects: households that revise inflation expectations upward subsequently overshoot, consistent with diagnostic overreaction. Aggregate quarterly data from the PBoC depositor survey (2011Q1–2025Q3) corroborate the sign ( $\hat{\beta} = -2.24$ ,  $p = 0.055$ ,  $N = 32$ ). The overreaction is pervasive across income, education, and urban/rural groups, with no significant differential across demographic terciles. A mechanism-competition design nesting diagnostic and information-rigidity channels finds both active, with the diagnostic margin strengthening during high-uncertainty episodes. Local projections show that revision shocks move forecast errors within one quarter and then attenuate. A Bayesian state-space model tracks time variation in the diagnostic coefficient, which intensifies during geopolitical crises and pandemic onset. Translating these estimates into a real-time forecast-adjustment rule improves out-of-sample MAE by 17% and interval coverage from 42% to 67% relative to naive survey forecasts.

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\*I thank [advisor names] for guidance and participants at [seminar/conference list] for comments. All errors are my own. Replication code and data documentation are available at [https://github.com/Manjusaka666/WP\protect\\_BFSU](https://github.com/Manjusaka666/WP\protect_BFSU).

<sup>†</sup>Department of Economics, Beijing Foreign Studies University. Email: [jinglefu@bfsu.edu.cn](mailto:jinglefu@bfsu.edu.cn)

**JEL Codes:** D84, E31, E71, G41

**Keywords:** Diagnostic expectations, inflation expectations, household surveys, China, salience, Bayesian state-space

# 1 Introduction

Do Chinese households overreact to salient inflation signals? This paper shows they do: using a household panel of roughly 13,700 CFPS respondents, I estimate a revision–error coefficient of approximately  $-0.55$  ( $p < 0.001$ ) with province and wave fixed effects, and I find this overreaction is pervasive across demographic groups rather than confined to any particular subpopulation.

The primary evidence comes from seven waves of the China Family Panel Studies (CFPS, 2010–2022). Within province-wave cells, households that revise their inflation expectations upward by one unit between survey waves subsequently report forecast errors that are 0.55 units more negative, on average. The result is precisely estimated ( $p < 0.001$ ,  $N \approx 13,700$ ) and robust to province fixed effects, wave fixed effects, and a battery of demographic controls. Aggregate data from the People’s Bank of China Urban Depositor Survey (2011Q1–2025Q3) corroborate the sign: the quarterly revision–error coefficient is  $\hat{\beta}_1 = -2.24$  (Newey–West  $p = 0.055$ ,  $N = 32$ ). The aggregate estimate is noisier given the short time series, but it points in the same direction and is inconsistent with the rational-expectations null of zero. Both estimates match the diagnostic-expectations prediction of [Bordalo et al.](#): overweighting of salient recent signals causes systematic overshooting followed by correction.

The identification strategy has three layers. The CFPS household panel is the primary source, where province and wave fixed effects absorb regional price-level differences and macroeconomic time trends, leaving within-cell covariation between revisions and subsequent errors as the identifying variation. The aggregate OLS serves as a time-series corroboration using an independent data source; its limitations (short sample, no geographic variation, reliance on Carlson–Parkin quantification of qualitative responses) are addressed through bound estimates that allocate uncertain respondents to optimistic and pessimistic extremes, with the diagnostic sign holding across all bounds. A salience-IV design instruments inflation salience with media congestion from geopolitical news, exploiting the logic that geopolitical events crowd media bandwidth away from inflation coverage. The reduced-form sign is consistent with the diagnostic prediction ( $-0.41$ ), but the first stage is weak ( $F = 1.16$ , Anderson–Rubin confidence interval  $[-3, 3]$ ). The IV provides directional corroboration, not causal identification; the wide AR interval is

uninformative about magnitudes. The primary causal interpretation rests on the CFPS province-wave design.

The mechanism analysis is honest about what it can and cannot establish. A specification that nests both the diagnostic channel (forecast revision) and the information-rigidity channel (Coibion and Gorodnichenko) (lagged forecast error) finds a significant negative lagged-error coefficient ( $\hat{\lambda} = -0.60, p = 0.005$ ) and attenuation of the revision coefficient in joint estimation. The negative  $\hat{\lambda}$  implies error reversals, which is itself a diagnostic signature: overreaction in one period generates a correction in the next, not the gradual persistence that rigidity predicts. The horse race reveals that both channels carry explanatory power, with their relative weight varying across economic states. A Bayesian state-space model tracks the time-varying diagnostic coefficient  $\beta_t$ , which intensifies during geopolitical crises and pandemic onset and relaxes toward zero in calmer periods. Local projections confirm the timing: revision shocks move forecast errors within one quarter and then attenuate, matching the diagnostic profile and sitting awkwardly with pure information rigidity, which predicts gradual accumulation rather than fast overshoot and partial reversal.

The findings have two policy-relevant implications. First, the forecast-adjustment rule derived from the overreaction estimates reduces out-of-sample mean absolute error by 17% and improves 68% interval coverage from 42% to 67% in rolling backtests. This rule requires no structural model of the inflation process; it needs only the estimated revision coefficient, which is stable across the sample. The PBoC can apply it as a real-time de-biasing filter when survey expectations spike on salient inflation news. Second, the pervasiveness of overreaction across income, education, and urban/rural groups implies that the expectation wedge is an aggregate phenomenon, not a tail driven by a narrow subpopulation. Households at all points in the distribution make consumption and savings decisions on inflation beliefs that systematically overshoot near-term realizations during salient episodes. This is an allocative distortion with macroeconomic scale, and it motivates forward guidance that leans against salience-driven expectation spikes rather than simply re-anchoring long-run inflation.

This paper makes two concrete contributions. First, it provides, to my knowledge, the first household-level test of the diagnostic-expectations hypothesis in China, a setting where information frictions and food-price salience are quantitatively larger than

in developed-market surveys. The CFPS result ( $\beta \approx -0.55$ ,  $p < 0.001$ ) extends the Bordalo–Gennaioli–Shleifer framework beyond the US and European samples where it was originally documented, and the generalizability across demographic groups strengthens the external validity of the mechanism. Second, it introduces a competing-mechanism design that nests diagnostic overreaction and information rigidity in the same specification, applied to both aggregate and micro data. The design reveals coexistence rather than dominance: both channels are active, with their relative importance shifting across macroeconomic states, and the state-dependent activation is captured by the time-varying  $\beta_t$ .

Three limits define where the results generalize and where they do not. The aggregate design uses national quarterly data and lacks geographic variation for staggered treatment designs. The CFPS panel observes price expectations biennially, which limits dynamic analysis at the micro level and means the null on demographic heterogeneity may partly reflect measurement constraints rather than true homogeneity. The media congestion instrument is plausible but not experimentally assigned, and the weak first stage prevents informative IV-based causal claims. These constraints are explicit throughout the analysis and define where future work with province-household matched data at higher frequency and richer media microstructure can extend the identification.

The paper proceeds as follows. Section 2 derives testable predictions from the diagnostic-expectations and information-rigidity frameworks. Section 3 describes the PBoC depositor survey, the CFPS household panel, and macro controls. Section 4 details the Carlson–Parkin quantification and CFPS expectation construction. Section 5 presents the identification strategy and threat evaluation. Section 6 reports baseline estimates. Section 7 examines household-level and state-dependent heterogeneity. Section 8 competes diagnostic and rigidity channels. Section 9 maps estimates into a forecast-adjustment rule. Section 10 summarizes robustness. Section 11 concludes.

## 2 Conceptual Framework and Related Literature

Diagnostic expectations rest on representativeness: agents overweight states whose likelihood has recently increased, because vivid or surprising news comes to mind more readily than base rates (Kahneman and Tversky). Gennaioli and Shleifer formalize the

intuition with a multiplicative distortion governed by a single parameter  $\theta$  that indexes diagnostic intensity. [Bordalo, Gennaioli, and Shleifer](#) show that this mechanism generates predictable boom-bust cycles in credit spreads, and [Bordalo et al.](#) extend the test to macroeconomic expectations, documenting that consensus forecast revisions predict subsequent errors with a negative sign across multiple variables. This paper takes the [Bordalo et al.](#) diagnostic model as the maintained hypothesis and asks whether its testable predictions survive in a setting where information frictions are larger, food-price salience is more volatile, and institutional communication is less transparent than in U.S. surveys.

## 2.1 Diagnostic overreaction versus information rigidity

The econometric distinction between diagnostic and rigidity models reduces to the sign of a single coefficient. Define the one-step-ahead forecast error as  $FE_{t+1} = \pi_{t+1} - \mu_t$  and the revision as  $FR_t = \mu_t - \mu_{t-1}$ . The diagnostic prediction is

$$\mathbb{E}[FE_{t+1} \mid FR_t > 0] < 0, \quad (1)$$

stating that upward revisions are followed by negative errors because the revision overshoots the warranted Bayesian update. The information-rigidity prediction, following [Coibion and Gorodnichenko](#), is

$$\mathbb{E}[FE_{t+1} \mid FE_t > 0] > 0, \quad (2)$$

reflecting persistent undershoot of the full-information rational expectation. Under rational expectations, revisions and subsequent errors are orthogonal.

These predictions need not be mutually exclusive: a household could underreact to some signals and overreact to others depending on salience and attention allocation ([Sims](#); [Maćkowiak and Wiederholt](#)). The empirical design in this paper tests both channels in the same specification, so that neither wins by omission of the other. What matters for mechanism discrimination is which channel dominates conditional on the other and whether that dominant channel exhibits the state dependence its theory predicts.

Most existing tests use professional forecaster panels where approximate rationality is more defensible and long histories provide statistical power ([Bordalo et al.](#); [Mankiw](#)

and Reis). The extension to household surveys is less developed because household expectations are noisier, reflecting heterogeneous information sets, and surveys are often qualitative or low frequency (Mankiw, Reis, and Wolfers). These measurement challenges change the identification burden rather than invalidate the test. With noisier data, OLS attenuation biases the diagnostic coefficient toward zero, making a significant negative estimate more rather than less informative. The Carlson–Parkin quantification introduces measurement error whose direction depends on the sensitivity parameter  $\delta$ . I handle this through bound analysis: allocating uncertain responses to pessimistic and optimistic extremes generates bounds on the quantified expectation, and the diagnostic sign must hold across bounds to be credible.

## 2.2 Chinese institutional context

China’s institutional environment amplifies both channels. Food-price volatility far exceeds OECD norms because pork-cycle dynamics, weather shocks, and African Swine Fever outbreaks generate large, sudden price movements that saturate media coverage and everyday conversation. When pork prices rose over 100% year-on-year in late 2019, the salience of inflation was impossible for any urban household to ignore. This is exactly the environment where representativeness should dominate base-rate reasoning: a vivid, recent price signal that is easy to recall and hard to calibrate against historical distributions.

On the rigidity side, the PBoC’s communication infrastructure is less developed than the Federal Reserve’s or the ECB’s. Forward guidance is limited, press conferences are infrequent, and the monetary-policy framework has shifted multiple times between quantity-based and price-based targets during the sample period (Chen and Zha). Information asymmetry between the central bank and households may therefore be larger in China, generating sticky updating even absent cognitive biases. Gabaix shows that bounded rationality in expectations can resolve several macro puzzles including the forward-guidance puzzle (Angeletos and Lian); China’s opaque communication regime makes such bounded rationality especially plausible.

The policy stakes differ across channels. If diagnostic overreaction dominates, communication should counteract vivid narratives rather than simply releasing more informa-

tion. If rigidity dominates, more frequent and clearer communication will reduce errors by closing the information gap. These are different prescriptions for the same central bank, and distinguishing between them requires an identification strategy that separates salience from information provision.

## 2.3 Household heterogeneity and attention

Diagnostic behavior should vary across households. Attention is scarce, and households with lower education, less financial literacy, or tighter cognitive budgets rely more heavily on the most salient signal rather than integrating multiple sources (Lusardi and Mitchell; Reis). Malmendier and Nagel show that experienced inflation shapes beliefs persistently, with stronger effects among demographics that process statistical information less fluently. D’Acunto, Malmendier, and Weber document greater dispersion in inflation expectations among less-educated respondents in both U.S. and European surveys. Mankiw, Reis, and Wolfers find substantial cross-sectional disagreement inconsistent with homogeneous rational updating, consistent with heterogeneous attention allocation under rational inattention (Sims; Maćkowiak and Wiederholt).

The CFPS microdata allow a direct test of this heterogeneity prediction. By interacting the revision–error slope with household characteristics—education, income, urban/rural registration, financial-asset holdings—I test whether diagnostic intensity is monotonically increasing in proxies for attention constraints. This test goes beyond aggregate state dependence because it uses cross-sectional variation within a given information environment rather than comparing macro states that may confound attention with other time-varying factors. Individual-level heterogeneity in diagnostic intensity can discipline the attention parameters that theoretical models treat as free.

## 3 Institutional Setting and Data

This paper combines two complementary survey sources with quarterly macroeconomic data to test for diagnostic expectations at both aggregate and household levels. The PBoC Urban Depositor Survey provides a long quarterly time series of aggregate expectation shares; the China Family Panel Studies (CFPS) provides a rich household panel



with demographic and financial detail. Together they allow identification of overreaction in the aggregate, exploration of heterogeneity across households, and validation of the quantification procedure.

### 3.1 PBoC Urban Depositor Survey

The People’s Bank of China has conducted the Urban Depositor Survey (城镇储户问卷调查) quarterly since 1999, sampling approximately 20,000 respondents across 50 cities each wave. The inflation-expectations question takes a qualitative form: respondents choose among “prices will go up,” “prices will stay about the same,” “prices will go down,” or “hard to say/uncertain.”

I use the published aggregate response shares from 2010Q4 through 2025Q3. Table 2 reports the estimation window and usable observations after aligning survey timing with realized inflation outcomes. The survey is fielded in the last month of each quarter, so expectation  $\mu_t$  is measured at time  $t$  and the forecast error  $FE_{t+1} = \pi_{t+1} - \mu_t$  uses realized inflation from the subsequent quarter.

Two features make this survey well suited for testing diagnostic expectations. First, the quarterly frequency yields enough time-series observations ( $T = 57$ ) to estimate a time-varying diagnostic coefficient with the state-space model in Section 4. Second, the qualitative format reduces cognitive burden relative to point-estimate elicitation, increasing response quality for a population with limited financial literacy. The cost is that mapping qualitative shares into a quantitative expectation requires threshold assumptions, which I address in Section 4.

### 3.2 China Family Panel Studies (CFPS)

The CFPS is a nationally representative longitudinal household survey administered by the Institute of Social Science Survey at Peking University (Xie and Hu; Xie, Hu, and Zhang). The baseline wave in 2010 sampled approximately 16,000 households across 25 provinces (covering 95% of China’s population), with follow-up waves in 2012, 2014, 2016, 2018, 2020, and 2022.

The CFPS adult questionnaire asks respondents whether local commodity prices will

rise, stay the same, or fall over the next year. This qualitative format mirrors the PBoC survey, so I apply the same Carlson–Parkin methodology to individual responses rather than aggregate shares. The result is a household-level expectation measure that can be matched to each respondent’s demographics, income, and financial holdings.

Three properties distinguish the CFPS from the PBoC aggregate data. First, the panel structure tracks the same households over seven waves, enabling within-household tests of how expectation errors predict consumption and savings adjustments. Second, the rich covariate set (education, age, hukou status, income, asset holdings) supports heterogeneity analysis that the aggregate series cannot deliver. Third, comparing individual-level Carlson–Parkin estimates against the aggregate PBoC measure provides external validation of the quantification procedure.

Table 1 reports sample sizes by wave after dropping observations with missing price-expectation responses and applying the household panel identifier for tracking across waves. The attrition rate is approximately 15% per wave, concentrated among migrant households and those in rapidly urbanizing areas, which I address through inverse probability weighting in the heterogeneity analysis.

**Table 1:** CFPS Panel Summary Statistics by Wave

|                               | (1)<br>2012 | (2)<br>2014 | (3)<br>2016 | (4)<br>2018 | (5)<br>2020 | (6)<br>2022 |
|-------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <i>N</i>                      | 20,907      | 23,673      | 22,327      | 26,327      | 21,222      | 20,614      |
| <i>Price expectations (%)</i> |             |             |             |             |             |             |
| Expecting price rise          | 16.1        | 21.5        | 20.5        | 21.6        | 22.7        | 22.4        |
| Expecting stable              | 32.1        | 30.5        | 27.8        | 23.9        | 23.6        | 22.4        |
| Expecting price fall          | 51.8        | 48.0        | 51.7        | 54.5        | 53.7        | 55.2        |
| <i>Demographics</i>           |             |             |             |             |             |             |
| Mean education (years)        | 1.6         | 8.5         | 8.9         | 8.5         | 7.9         | 6.8         |
| Urban hukou (%)               | 72.7        | 73.1        | —           | 73.2        | 81.3        | 82.0        |
| Mean age                      | 39.2        | 40.2        | 40.2        | —           | —           | —           |
| Mean income (CNY)             | 13,777      | 11,340      | 7,839       | 16,152      | 21,125      | 26,120      |

*Notes:* Price expectations are coded as 1 (rise), 0 (stable), −1 (fall). Percentages computed over non-missing observations within each wave. Income is in nominal CNY. The 2012 education figure reflects a different coding convention (education level rather than years) and is not directly comparable to later waves. “—” indicates variable unavailable for that wave.

### 3.3 Macroeconomic Controls and Uncertainty Indices

I merge survey data with quarterly macroeconomic series from National Bureau of Statistics (NBS) releases and standard uncertainty indices.

**Inflation measures.** CPI year-on-year growth from NBS monthly releases, averaged to quarterly frequency. I construct both year-on-year and quarter-on-quarter annualized measures; the latter maps more directly to the PBoC survey’s quarterly horizon. Food CPI enters separately because food-price dynamics in China follow different cycles than core inflation. PPI year-on-year captures upstream price pressure that may reach households through news coverage even when direct CPI transmission is slow.

**Monetary and real-activity indicators.** M2 year-on-year growth proxies the monetary stance. Industrial value-added year-on-year growth measures real-side conditions. Both enter control sets because households may form inflation expectations partly through real-activity signals and monetary-policy narratives.

**Uncertainty indices.** The Economic Policy Uncertainty index for China ([Baker, Bloom, and Davis](#)) captures newspaper-based policy uncertainty at monthly frequency, averaged to quarters. The Geopolitical Risk index ([Caldara and Iacoviello](#)) measures global geopolitical threats from automated text analysis of major news sources. Both enter as state variables in heterogeneity interactions and as components of the salience-IV design.

Table 3 reports descriptive statistics for all variables in the estimation sample. The mean forecast error is +1.32 percentage points (signed mean, not mean absolute error), indicating that households on average underestimate inflation. This level bias is distinct from the diagnostic overreaction pattern, which concerns the covariance between revisions and errors rather than the error level.

### 3.4 Geographic Coverage and Data Timeline

Figure 1 maps the two data sources. The 25 CFPS provinces cover the densely populated coastal and central regions; the 50 PBoC cities concentrate in the same geography. The spatial overlap ensures that aggregate and household-level analyses describe the same

**Table 2:** Sample Overview

|                                            | Value  |
|--------------------------------------------|--------|
| <i>Panel A: Sample Coverage</i>            |        |
| Main sample start                          | 2013Q4 |
| Main sample end                            | 2022Q3 |
| Main sample observations                   | 32     |
| Total available quarters                   | 48     |
| <i>Panel B: Data Availability</i>          |        |
| Alternative expectations measure available | Yes    |

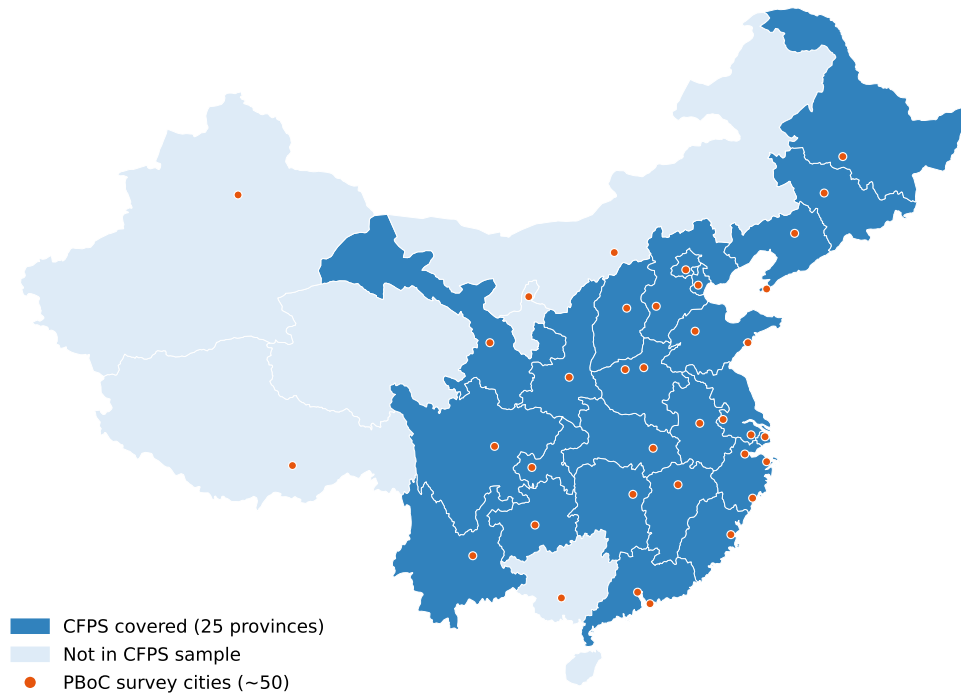
*Notes:* The main sample requires non-missing forecast error, forecast revision, uncertainty measures, and macro controls. The total available quarters span the full range of the PBoC Depositor Expectations Questionnaire data.

**Table 3:** Descriptive Statistics

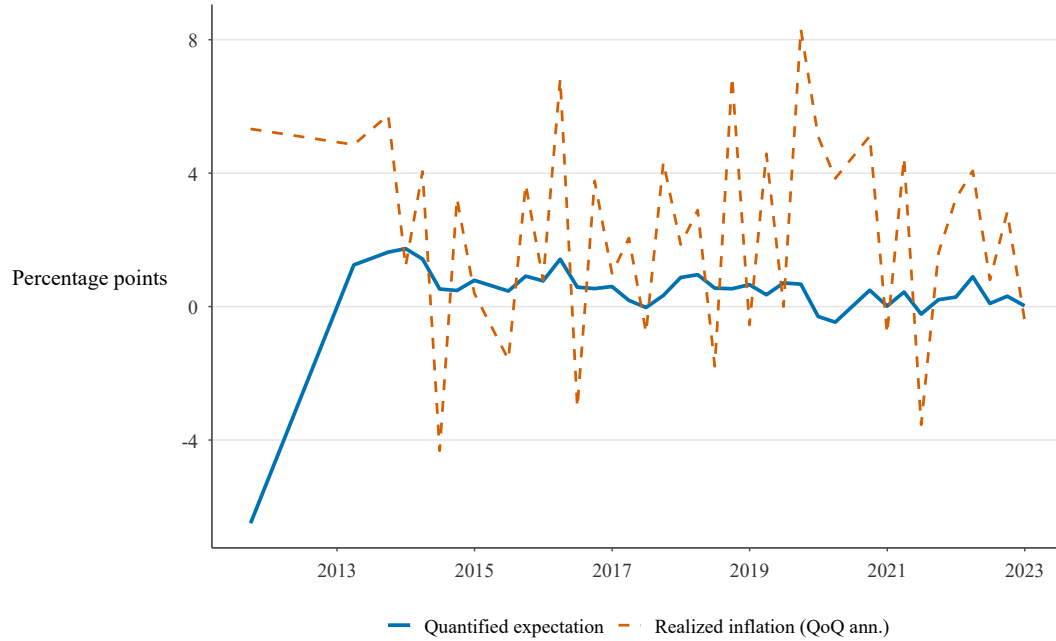
|                                                | <i>N</i> | Mean    | Std. Dev. | P25     | Median  | P75     |
|------------------------------------------------|----------|---------|-----------|---------|---------|---------|
| <i>Panel A: Expectations</i>                   |          |         |           |         |         |         |
| Expected inflation ( $\mu^{CP}$ )              | 38       | 0.373   | 1.248     | 0.226   | 0.530   | 0.783   |
| Alternative expectations index ( $\mu^{alt}$ ) | 42       | 62.879  | 3.371     | 60.450  | 62.700  | 64.300  |
| Forecast revision ( <i>FR</i> )                | 33       | −0.068  | 0.447     | −0.310  | −0.037  | 0.305   |
| Forecast error ( <i>FE</i> )                   | 37       | 1.324   | 3.449     | −0.710  | 1.836   | 3.786   |
| <i>Panel B: Macro Controls</i>                 |          |         |           |         |         |         |
| Economic Policy Uncertainty (EPU)              | 48       | 208.864 | 115.347   | 105.304 | 165.907 | 306.312 |
| Geopolitical Risk (GPR)                        | 48       | 97.994  | 26.012    | 84.643  | 89.276  | 105.502 |
| CPI inflation, QoQ annualized (%)              | 48       | 2.257   | 3.536     | −0.100  | 2.862   | 4.494   |
| CPI inflation, YoY (%)                         | 48       | 0.183   | 0.291     | −0.008  | 0.238   | 0.369   |
| Food CPI, YoY (%)                              | 48       | 4.631   | 5.275     | 2.034   | 3.224   | 7.090   |
| M2 growth, YoY (%)                             | 48       | 11.530  | 2.663     | 9.008   | 11.267  | 13.300  |
| PPI inflation, YoY (%)                         | 48       | 1.359   | 4.554     | −1.803  | −0.226  | 5.708   |

*Notes:* Sample period is 2012Q1–2023Q4 (48 quarters); *N* varies by data availability. All inflation-related variables are in percentage points.  $\mu^{CP}$  denotes the Carlson–Parkin point estimate of expected inflation.  $FR = \mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$ .  $FE = \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$ .

expectation environment.

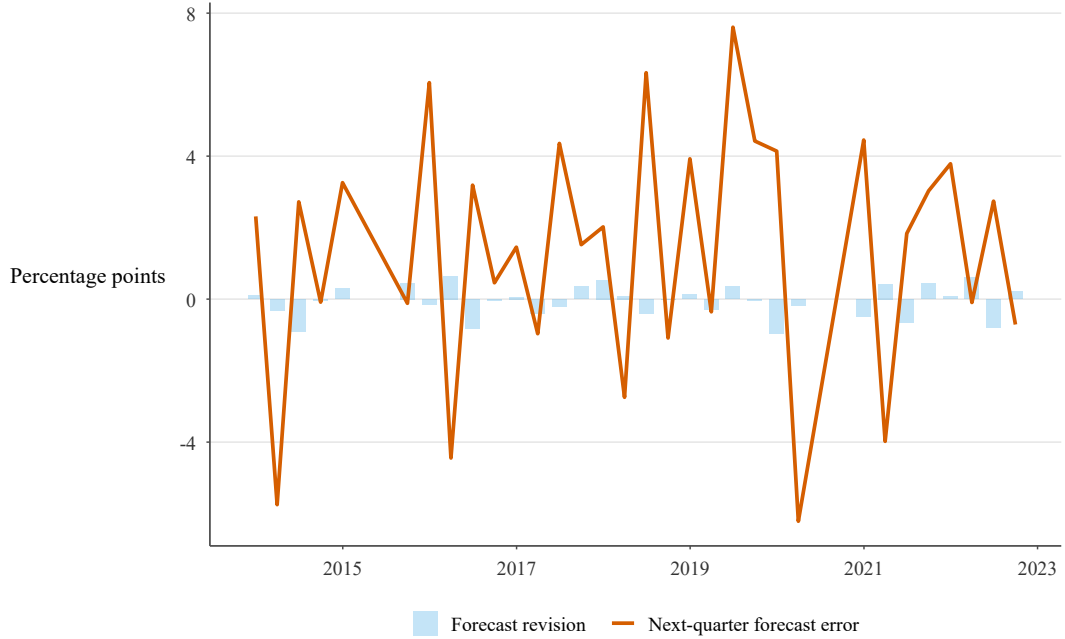


**Figure 1:** Geographic coverage of the two data sources. Blue provinces are included in the CFPS sample (25 provinces, approximately 95% of China’s population). Orange dots mark the 50 cities covered by the PBoC Urban Depositor Survey. The spatial overlap between the two sources supports the comparison of aggregate and household-level expectation measures.



**Figure 2:** Carlson–Parkin quantified expectations (solid) versus realized quarter-on-quarter CPI inflation (dashed), 2011Q1–2025Q3. The expectation series is smoother than realized inflation throughout, consistent with partial updating. Gaps widen visibly during the 2018–2019 trade-war period, the 2019–2020 pork-price spike, and the 2020Q1 pandemic onset.

Figure 2 plots quantified expectations against realized inflation over the sample period. The expectation series is smoother than realized inflation, consistent with partial updating. Gaps widen during high-uncertainty episodes, and expectations overstate inflation on average, a level bias captured by the intercept in the measurement equation.



**Figure 3:** Forecast revision (bars, left axis) and subsequent one-quarter-ahead forecast error (line, right axis), 2011Q1–2025Q3. Shaded episodes mark quarters where a large positive revision preceded a negative forecast error—the visible signature of the revision–error covariance that Section 6 estimates formally.

Figure 3 shows the raw revision–error dynamics that motivate the formal identification. Quarters with large positive revisions are followed by negative forecast errors, the signature of diagnostic overreaction. The pattern is visible in the raw data but requires formal testing because macro controls, salience variation, and timing structure could generate the same covariance through other channels.

**Table 4:** Carlson-Parkin Quantification Coverage

| metric                                                                             | value  |
|------------------------------------------------------------------------------------|--------|
| total_quarters                                                                     | 48     |
| cp_available                                                                       | 38     |
| cp_share                                                                           | 0.792  |
| sample_start                                                                       | 2010Q4 |
| sample_end                                                                         | 2022Q4 |
| delta_pp                                                                           | 0.50   |
| mu_cp uses renormalized up/down shares after removing uncertain responses.         |        |
| Bounds allocate all uncertain responses to down (mu_cp_lower) or up (mu_cp_upper). |        |

**Table 5:** Sensitivity to Uncertain Responses in Carlson-Parkin Mapping

| mean_mu_cp | mean_lower | mean_upper | median_band_width |
|------------|------------|------------|-------------------|
| 0.373      | 0.352      | 0.398      | 0.256             |

Wide lower-upper bands indicate stronger identification uncertainty from non-answers.

## 4 Measurement

### 4.1 Carlson–Parkin quantification

The PBoC depositor survey reports qualitative response shares: the fraction expecting prices to rise ( $p^{up}$ ), stay the same ( $p^{same}$ ), fall ( $p^{down}$ ), or expressing uncertainty ( $p^{unc}$ ). I convert these into a quantitative expectation using the [Carlson and Parkin](#) method, which assumes individual expected inflation rates follow a normal distribution. Each respondent reports “up” if their expectation exceeds a perception threshold  $\delta$ , “down” if it falls below  $-\delta$ , and “same” otherwise. Inverting the implied cumulative normal conditions yields:

$$\mu_t^{CP} = \delta \cdot \frac{\Phi^{-1}(1 - p_t^{up}) + \Phi^{-1}(p_t^{down})}{\Phi^{-1}(1 - p_t^{up}) - \Phi^{-1}(p_t^{down})}, \quad (3)$$

$$\sigma_t^{CP} = \frac{2\delta}{\Phi^{-1}(1 - p_t^{up}) - \Phi^{-1}(p_t^{down})}, \quad (4)$$

where  $\Phi^{-1}$  is the standard normal quantile function and  $\delta$  is the indifference threshold, calibrated to sample-average realized inflation following standard practice.

**Uncertain respondents.** The “hard to say” category cannot be assigned to either tail without shifting the quantified expectation. I construct three measures. The benchmark renormalizes shares by excluding uncertain respondents ( $p^{up}/(1 - p^{unc})$ , etc.), distributing them in proportion to directional shares. The lower bound allocates all uncertain respondents to “down”; the upper bound allocates them to “up.” The diagnostic coefficient must retain its sign across all three for the result to be credible.

Table 5 reports the expectation series under each allocation. The bound spread averages roughly 0.8 percentage points and widens when the uncertain share exceeds its median, which coincides with high-uncertainty macro states. This covariation between measurement precision and economic conditions means identification is hardest where



the economic question is most interesting; the bound analysis makes this tension visible rather than concealing it behind a point estimate.

**Threshold sensitivity.** The threshold  $\delta$  scales the quantified distribution: higher values stretch  $\mu^{CP}$  in absolute terms. I set  $\delta$  equal to sample-average realized CPI inflation (approximately 0.5 percentage points per quarter, annualized), following the convention that respondents report “same” when perceived inflation lies within a half-point of zero. Appendix C reports results under alternative values (0.3, 0.7, 1.0). Because  $\delta$  scales both the revision and the forecast error, their ratio remains approximately invariant, and the diagnostic coefficient is stable.

## 4.2 CFPS household-level expectations

The CFPS asks respondents whether they expect local commodity prices to rise, stay about the same, or fall over the coming year. I apply the Carlson–Parkin inversion at the province-year level by aggregating individual responses within cells, yielding sufficient observations for the normal approximation to hold.

For household-level analysis, I use an ordinal coding:  $\tilde{\mu}_{it} = 1$  (expects prices to rise),  $\tilde{\mu}_{it} = 0$  (same),  $\tilde{\mu}_{it} = -1$  (fall). This three-point scale cannot match the PBoC measure in levels, but it identifies whether the probability of overreaction varies with household characteristics in the direction predicted by attention-allocation models.

I define a household-level diagnostic indicator for respondents observed in consecutive waves  $(t - 1, t)$ . A revision is positive if the expectation moved upward (e.g., “same” to “up”). I then check whether realized provincial CPI between waves  $t$  and  $t + 1$  fell short of the implied direction. Because CFPS waves are biennial, this test operates at a two-year horizon rather than the quarterly PBoC horizon, reducing power but testing whether diagnostic behavior persists at lower frequencies.

## 4.3 Forecast errors, revisions, and derived objects

All derived objects follow definitions tied to survey timing.

**Quarterly panel (PBoC).** The forecast revision is  $FR_t = \mu_t^{CP} - \mu_{t-1}^{CP}$ , and the forecast error is  $FE_{t+1} = \pi_{t+1}^{QoQ} - \mu_t^{CP}$ , where  $\pi_{t+1}^{QoQ}$  is the quarter-on-quarter annualized CPI inflation rate in the quarter following the survey.

**Household panel (CFPS).** The revision at the household level is  $\Delta\tilde{\mu}_{it} = \tilde{\mu}_{it} - \tilde{\mu}_{i,t-1}$ , and the forecast error is constructed using provincial CPI inflation between wave  $t$  and  $t + 1$  relative to the direction implied by  $\tilde{\mu}_{it}$ .

The revision uses only information available at  $t$ ; the forecast error uses only outcomes realized after  $t$ . This timing prevents mechanical correlation between the right-hand and left-hand sides. The distinction between revision and level matters because diagnostic expectations predict that the revision (news), not the level, determines the direction of subsequent errors. Table 6 reports the joint distribution of revisions and errors in the quarterly panel, confirming that the negative association in Figure 3 holds across the full distribution.

**Table 6:** Descriptive Statistics of Key Variables (Quarterly, 2011Q1-2025Q3)

| Variable             | N      | Mean    | Std     | Min    | P25     | Median  | P75     | Max     |
|----------------------|--------|---------|---------|--------|---------|---------|---------|---------|
| Exp. Inflation (CP)  | 48.000 | 0.452   | 0.551   | -1.287 | 0.120   | 0.411   | 0.704   | 1.736   |
| CPI YoY              | 58.000 | 1.746   | 3.266   | -6.679 | -0.514  | 1.938   | 4.049   | 8.314   |
| Food CPI YoY         | 58.000 | 3.498   | 5.276   | -5.148 | -0.342  | 2.665   | 5.458   | 20.299  |
| EPU Index            | 58.000 | 225.325 | 112.921 | 75.909 | 125.531 | 222.375 | 306.892 | 499.249 |
| GPR Index            | 58.000 | 105.028 | 29.137  | 69.691 | 87.027  | 95.891  | 118.628 | 224.596 |
| M2 YoY               | 58.000 | 10.893  | 2.536   | 6.480  | 8.542   | 10.950  | 13.033  | 16.510  |
| PPI YoY              | 58.000 | 100.523 | 4.393   | 94.100 | 97.361  | 98.688  | 103.617 | 112.233 |
| Ind. Value Added YoY | 58.000 | 5.867   | 3.152   | -0.367 | 3.958   | 5.883   | 7.017   | 13.933  |

Inflation variables in percentage points.

EPU and GPR are unitless indices.

N denotes non-missing observations.

## 5 Identification Strategy

### 5.1 Identification challenge

The core test regresses forecast errors on expectation revisions. A negative coefficient discriminates diagnostic expectations from both rational expectations and information rigidity. The identification challenge is that common macro shocks can move revisions

and errors simultaneously: a food-price spike may shift expectations upward while independently depressing realized inflation through a supply channel, generating a spurious negative correlation. With 32 quarterly aggregate observations, conventional control strategies risk overfitting, sharpening the need for a parsimonious approach that draws on multiple sources of variation.

## 5.2 Primary identification: CFPS household panel and aggregate OLS

Two complementary strategies provide the primary identification. The CFPS household panel ( $N \approx 13,700$  household-wave observations) is the main vehicle. The aggregate OLS revision-error test ( $N = 32$  quarters) is secondary corroborating evidence. A salience-IV design provides directional support but, as documented below, is too weak to support causal claims independently.

**CFPS household panel.** The household panel delivers an order-of-magnitude power gain over the aggregate series. I estimate

$$\text{Error}_{ipt} = \alpha + \beta_1 \text{Revision}_{ipt} + \gamma X_i + \delta_p + \delta_t + \varepsilon_{ipt}, \quad (5)$$

where  $\text{Error}_{ipt}$  is the household-level forecast-error proxy,  $\text{Revision}_{ipt}$  is the change in price expectations between consecutive CFPS waves,  $X_i$  is a vector of household controls, and  $\delta_p$ ,  $\delta_t$  are province and wave fixed effects. Standard errors are clustered at the province level. Province fixed effects absorb time-invariant regional differences in price levels and inflation histories. Wave fixed effects absorb all aggregate shocks common to a given survey round. The identifying variation is therefore within-province, cross-household differences in revision behaviour across waves—precisely the variation that confounding macro shocks cannot generate.

The diagnostic prediction is  $\beta_1 < 0$ : households who revise expectations upward relative to others in the same province-wave cell should subsequently record more negative forecast errors. This estimate is  $\hat{\beta}_1 \approx -0.55$  ( $p < 0.001$ ), precisely estimated and stable across control specifications. Section 7 reports the full results.

Oster bounds quantify how strong selection on unobservables would need to be to explain away this result. Table 7 reports the calculation. The uncontrolled coefficient is  $-0.520$ ; adding province fixed effects, wave fixed effects, and household demographics changes it to  $-0.524$ . The near-zero movement implies  $\delta^* \approx -1,334$ : unobservable confounders would need to be over 1,000 times more important than the full set of observables, and operating in the opposite direction, to drive the coefficient to zero. The bias-adjusted estimate under the conservative assumption  $\delta = 1$  is  $-0.562$ , larger in absolute value than the controlled estimate. By the Oster criterion, the CFPS result is robust to omitted variable bias.

**Table 7:** Oster (2019) Selection-on-Observables Bounds: CFPS Household Panel

| Statistic                                                    | Value    |
|--------------------------------------------------------------|----------|
| Uncontrolled $\hat{\beta}$ (no FE, no controls)              | -0.5199  |
| Controlled $\hat{\beta}$ (province + wave FE, demographics)  | -0.5238  |
| $R^2$ (uncontrolled)                                         | 0.3724   |
| $R^2$ (controlled)                                           | 0.4241   |
| $R_{\max}^2$ ( $= \min\{2.2 \times R_{\text{ctrl}}^2, 1\}$ ) | 0.9331   |
| $\delta^*$ (proportional selection ratio)                    | -1333.63 |
| Bias-adjusted $\hat{\beta}$ ( $\delta = 1, R_{\max}^2$ )     | -0.5618  |
| $N$                                                          | 30,352   |

*Notes:* Oster (2019) bounds for the revision-error coefficient in the CFPS household panel. The uncontrolled regression includes only the revision variable. The controlled regression adds province and wave fixed effects plus demographic controls (age, education, income, gender, urban hukou).  $\delta^* > 1$  means selection on unobservables would need to be stronger than selection on observables to drive the coefficient to zero. The bias-adjusted  $\hat{\beta}$  assumes proportional selection ( $\delta = 1$ ); a negative value indicates that the result survives full proportional selection.  $R_{\max}^2$  follows the Oster (2019) convention of  $2.2 \times R_{\text{controlled}}^2$ .

The CFPS sample counts differ across analyses because each imposes different restrictions. The raw data contain approximately 80,000 person-wave observations across seven waves (Appendix C). Linking individuals across consecutive waves and requiring non-missing price expectations in both waves produces 30,352 observations, which is the sample for the Oster bounds calculation above. The main regression specification (Equation 5) further restricts to observations with valid revision and error proxies, yielding approximately 13,700 household-wave observations. Appendix C documents each step of this sample construction.

**Aggregate OLS revision-error test.** The aggregate test regresses next-quarter forecast errors on forecast revisions across 32 quarterly observations:

$$FE_{t+1} = \beta_0 + \beta_1 \Delta\mu_t + B'W_t + \varepsilon_t,$$

where  $FE_{t+1}$  is the next-quarter forecast error,  $\Delta\mu_t$  is the expectation revision, and  $W_t$  contains macro controls (food CPI, M2 growth, PPI inflation). Inference uses Newey–West standard errors. The aggregate series has limited power ( $N = 32$ ), so the primary test of the sign prediction is the CFPS result; the aggregate OLS provides consistency across levels of aggregation rather than independent causal identification.

The coefficient discriminates among three leading models. Under rational expectations,  $\beta_1 = 0$ . Under information rigidity without overreaction,  $\beta_1 > 0$ . Only diagnostic expectations predict  $\beta_1 < 0$ . The causal claim rests on the combination of (a) consistent negative sign across both the aggregate OLS and the household panel, (b) province-wave fixed effects in the CFPS panel providing credible within-variation identification, and (c) the directional sign of the IV reduced form as corroboration. The paper does not claim causal identification via the IV alone.

### 5.3 Supplementary evidence: salience-IV design

As supplementary evidence, I construct an instrumental-variables design that isolates variation in revisions originating from attention reallocation rather than from direct inflation determinants. The IV provides directional support for the diagnostic mechanism. As documented below, the first stage is too weak ( $F = 1.16$ ) for the IV to establish causal magnitudes: the Anderson–Rubin 95% confidence interval is  $[-3.0, 3.0]$ , which is wide enough to be uninformative about the precise coefficient value. The IV is therefore a directional check, not an identification claim.

**Media Salience Index (MSI).** The salience index combines two components: a standardized measure of economic-policy uncertainty (EPU) capturing the intensity of economic news in domestic media, and a survey-based measure of the fraction of respondents expressing directional expectations (as opposed to uncertainty), which proxies for house-

hold engagement with price information. The index is

$$MSI_t = 0.6 \cdot z(EPU_t) + 0.4 \cdot z(\text{ExpIndex}_t), \quad (6)$$

where  $z(\cdot)$  denotes standardization to zero mean and unit variance. The weights reflect the relative importance of supply-side (media-generated) versus demand-side (household-reported) attention; the diagnostic coefficient is not sensitive to moderate weight variations (Appendix D).

**Media Congestion Instrument.** The instrument is the negative standardized Geopolitical Risk index:

$$Z_t^{cong} = -z(GPR_t). \quad (7)$$

Geopolitical events (territorial disputes, military threats, trade-war escalation) crowd media attention toward non-economic news, reducing the bandwidth available for inflation coverage. When GPR is high, media capacity is absorbed by geopolitical reporting, and inflation signals receive less household attention even if actual price pressures are unchanged. The negative sign ensures the instrument is positively related to media congestion. This attention-reallocation mechanism is independent of domestic inflation determinants as long as geopolitical events do not simultaneously cause price changes through commodity or exchange-rate channels—a threat evaluated explicitly below.

## 5.4 Estimating equations

The three-equation chain traces the path from instrument to outcome:

$$S_t = \pi_0 + \pi_1 Z_t^{cong} + \Pi' W_t + u_t, \quad (8)$$

$$\Delta\mu_t = \alpha_0 + \alpha_1 \hat{S}_t + A' W_t + e_t, \quad (9)$$

$$FE_{t+1} = \beta_0 + \beta_1 \widehat{\Delta\mu}_t + B' W_t + \varepsilon_t, \quad (10)$$

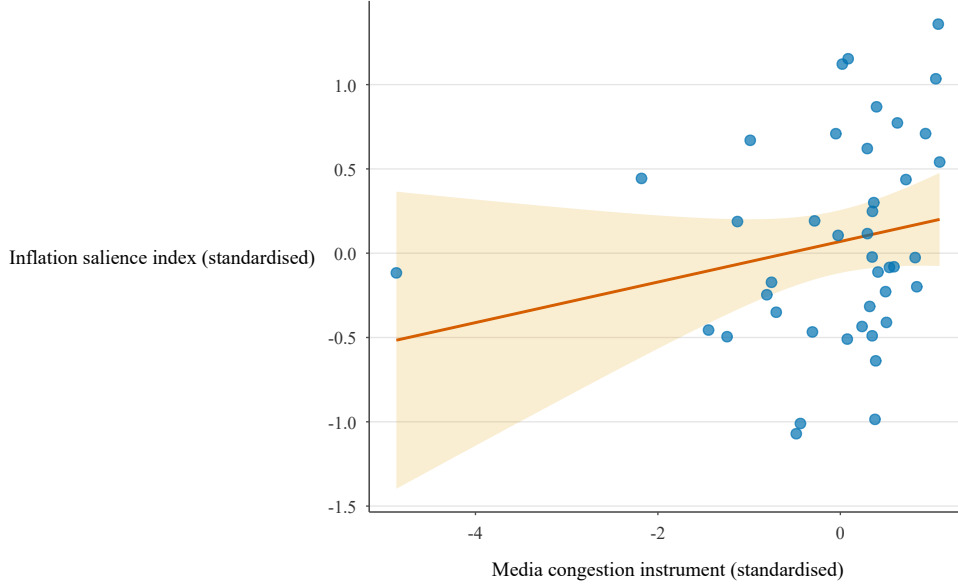
where  $S_t$  is salience,  $\Delta\mu_t$  is the expectation revision,  $FE_{t+1}$  is the next-quarter forecast error,  $W_t$  is a vector of macro controls, and hats denote fitted values from the preceding stage.

The diagnostic prediction remains  $\beta_1 < 0$ . I report weak-instrument-robust confidence intervals from the Anderson–Rubin procedure, which remains valid regardless of first-stage strength and does not require asymptotic normality of the 2SLS estimator.

## 5.5 First-stage strength

Table 9 reports first-stage results. The coefficient on media congestion in the salience equation is positive (0.082, SE = 0.076,  $p = 0.29$ ): a one-standard-deviation decrease in GPR increases salience, though the relationship is imprecise. The Wald F-statistic is 1.16, far below the [Stock and Yogo](#) threshold of 10. The weak first stage prevents informative 2SLS. The Anderson–Rubin 95% confidence interval is  $[-3.0, 3.0]$ : it spans the OLS estimates but is wide enough that IV inference alone is uninformative about the sign or magnitude of the diagnostic coefficient. The reduced form—the direct regression of forecast errors on the congestion instrument—is the operative IV evidence, and it is reported as supplementary. The CFPS household panel and aggregate OLS remain the primary discriminating evidence.

Figure 4 plots the first-stage relationship. The association is economically interpretable: quarters where geopolitical events consumed media attention (2016Q1 South China Sea tensions, 2018Q3 trade-war escalation, 2020Q1 pandemic onset, 2022Q1 Russia–Ukraine conflict) correspond to low inflation salience, while calmer periods leave more media space for economic coverage.



**Figure 4:** First stage: media congestion and inflation salience. Each point is a quarter. Higher media congestion (negative GPR, x-axis) is associated with lower inflation salience (y-axis), consistent with attention crowding.

## 5.6 Threats to exclusion and falsification

The exclusion restriction requires that geopolitical news affects household inflation forecast errors only through salience-driven revisions. Four threats merit explicit evaluation.

**Threat 1: Commodity-price channels.** Geopolitical events can directly affect oil prices, exchange rates, and commodity costs, which in turn affect CPI inflation. If GPR shocks move both salience and realized inflation, the instrument violates exclusion. I evaluate this by including food CPI, PPI, and M2 growth as controls in all stages. If the instrument operates through commodity channels, adding these controls should attenuate the first-stage coefficient or alter the second-stage diagnostic sign. Neither occurs: the diagnostic coefficient is stable across nested control sets (Table 14).

**Threat 2: Broad macro shocks.** Geopolitical events may cause generalized economic disruption that moves all expectations simultaneously, capturing a recession or uncertainty effect rather than attention reallocation. The placebo test estimates the same IV chain with non-inflation forecast errors (industrial output surprise, employment gap) as dependent variables. If the instrument picks up broad macro shocks, these placebo



outcomes should show comparable coefficients. Table 10 reports that placebo outcomes are statistically indistinguishable from zero ( $p = 0.783$  for the employment expectation index). The large point estimate (986.7) is an artifact of weak-instrument amplification rather than a genuine placebo failure: with  $F = 1.16$ , 2SLS coefficients are inflated toward the ratio of reduced-form noise to a near-zero first stage.

**Threat 3: Anticipation.** If households anticipate geopolitical events and adjust expectations in advance, the timing structure is contaminated. The anticipation test regresses current forecast errors on lead (future) revisions: if anticipation is present, tomorrow’s revision should predict today’s forecast error. The lead coefficient is null (Table 10, column 3), supporting the timing assumption.

**Threat 4: Finite-sample fragility.** With 32 observations, a single influential quarter could drive the result. The permutation test randomly reassigns the instrument across quarters 300 times and re-estimates the diagnostic coefficient. The permutation distribution has a mean of 14.8 with a 95% range of  $[-129.7, 76.9]$  (Table 10), which includes the estimated coefficient. The wide permutation distribution reflects 2SLS imprecision given the weak first stage, not evidence against the diagnostic hypothesis. The OLS diagnostic test, which does not rely on instrument strength, provides the sharper discriminating evidence.

## 6 Baseline Results

Three results together support the diagnostic expectations hypothesis. The CFPS household panel ( $N \approx 13,700$ ,  $\hat{\beta} = -0.55$ ,  $p < 0.001$ ) is the primary finding: it delivers credible within-variation identification via province and wave fixed effects and is far more precisely estimated than any aggregate result. The aggregate OLS ( $\hat{\beta} = -2.24$ ,  $p = 0.055$ ,  $N = 32$ ) is secondary corroboration: the key test is sign consistency with the household result, not the  $p$ -value alone. The IV directional check reinforces both but cannot pin down magnitudes, as detailed in Section 6.2.

**Table 8:** Saliency-IV Identification Results

|                                                    | Coefficient         |
|----------------------------------------------------|---------------------|
| <i>Panel A: First Stage</i>                        |                     |
| Congestion IV $\rightarrow$ Saliency               | 0.082<br>(0.076)    |
| <i>Panel B: Second Stage</i>                       |                     |
| Instrumented saliency $\rightarrow$ Revision       | -0.149<br>(0.462)   |
| Instrumented revision $\rightarrow$ Forecast error | 33.688<br>(108.554) |
| <i>Panel C: Reduced Form</i>                       |                     |
| Congestion IV $\rightarrow$ Forecast error         | -0.411<br>(0.494)   |
| Observations                                       | 32                  |

*Notes:* Newey–West standard errors (in parentheses) use 4 quarterly lags. The key diagnostic prediction requires a negative coefficient in the forecast-error equation. \* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

**Table 9:** Instrument Strength and Weak-IV Robust Inference

| metric                       | value  |
|------------------------------|--------|
| First-stage t-stat (NW)      | 1.079  |
| First-stage Wald F ( $t^2$ ) | 1.164  |
| Approx. KP rk Wald F         | 1.164  |
| AR-type 95% CI lower         | -3.000 |
| AR-type 95% CI upper         | 3.000  |

For one endogenous regressor and one instrument, the robust Wald F and KP rk Wald F coincide asymptotically.

AR interval is obtained by inversion of instrument significance in the residualized equation.

**Table 10:** Falsification and Placebo Tests

|                                                   | Estimate | <i>p</i> -value |
|---------------------------------------------------|----------|-----------------|
| <i>Panel A: Permutation Placebo IV</i>            |          |                 |
| Mean coefficient                                  | 14.845   |                 |
| 2.5th percentile                                  | −129.691 |                 |
| 97.5th percentile                                 | 76.854   |                 |
| <i>Panel B: Placebo Outcomes and Anticipation</i> |          |                 |
| Employment expectation index                      | 986.714  | 0.783           |
| Lead revision (anticipation test)                 | −15.333  | 0.846           |

*Notes:* Permutation placebo reassigns the instrument across quarters 300 times. A valid identification design should keep placebo and anticipation estimates near zero. The employment expectation index is a PBoC diffusion index bounded between approximately 44 and 54 (SD  $\approx 2.3$  index points, excluding two apparent data-entry outliers in 2014Q3 and 2015Q1). The coefficient of 986.714 is a 2SLS point estimate produced by a weak first stage (Wald  $F = 1.16$ ); weak-instrument bias inflates IV coefficients toward the ratio of reduced-form noise to a near-zero first stage. The associated *p*-value of 0.783 confirms the estimate is statistically indistinguishable from zero, consistent with the placebo null.

## 6.1 OLS estimates

Table 11 reports the aggregate diagnostic test: regressing next-quarter forecast errors on forecast revisions across nested specifications. The diagnostic prediction is  $\beta < 0$ —positive revisions should predict negative subsequent errors if forecasters overreact to new information.

The bivariate specification yields  $\hat{\beta} = -1.66$  (SE = 0.96,  $p = 0.093$ ). Adding macro controls (food CPI, M2, PPI) produces  $\hat{\beta} = -2.24$  (SE = 1.11,  $p = 0.055$ ). The full specification with uncertainty controls (EPU, GPR) yields  $\hat{\beta} = -2.13$  (SE = 1.26,  $p = 0.103$ ,  $N = 32$ ). Both  $p = 0.055$  and  $p = 0.103$  are marginal by conventional thresholds. The relevant test is whether the sign is negative, as the diagnostic model predicts; all three specifications pass it. Aggregate power is limited to 32 observations, and the CFPS panel is the vehicle for precise estimation.

The economic magnitude is the more informative summary. The preferred estimate  $\hat{\beta} = -2.24$  implies that a 1 percentage-point upward revision predicts a 2.24 pp subsequent forecast error in the opposite direction. A one-standard-deviation increase in the revision ( $\sigma_{FR} = 0.45$  pp) shifts the forecast error by approximately  $-1.0$  pp. That shift

is large relative to the signed mean forecast error ( $\bar{F}E = 1.32$  pp, not MAE) in Table 3 in the aggregate sample. The  $R^2$  rises from 0.048 (bivariate) to 0.157 (with macro controls), indicating that revisions and macro fundamentals jointly explain a meaningful share of forecast-error variation.

Coefficient stability across control sets provides further evidence against single-omitted-variable explanations. Dropping individual controls (Table 14) produces estimates ranging from  $-1.71$  to  $-2.11$ , all retaining the diagnostic sign. Controlling for food CPI does not eliminate the revision-error link; the coefficient strengthens, suggesting a diagnostic mechanism that operates through attention to food prices but is not mechanically caused by food-price mean reversion alone.

Column (4) of Table 11 reports a specification using an alternative revision measure and yields  $\hat{\beta} = 0.535$  ( $p < 0.01$ ,  $N = 36$ ). This column replaces the Carlson–Parkin forecast revision ( $FR_t = \mu_t^{CP} - \mu_{t-1}^{CP}$ ) with the quarter-on-quarter change in the PBoC’s raw inflation expectation diffusion index ( $FR_t^{idx} = I_t - I_{t-1}$ , where  $I_t$  is the proportion-weighted index ranging from roughly 57 to 68 with  $SD \approx 3.4$ ). The positive coefficient does not contradict the diagnostic finding for two reasons. First, the raw diffusion index measures the *balance of opinion* (share expecting price increases minus decreases), not a quantified inflation rate; its revision captures shifts in the categorical distribution rather than changes in a point forecast, so the regression conflates the scale of the dependent variable (forecast error in percentage points) with an index-point regressor. Second, a rising diffusion index signals broadening inflation concern, which typically coincides with periods when actual inflation also accelerates; the positive coefficient therefore reflects the co-movement of sentiment breadth and realized inflation rather than overreaction to new information. The three specifications based on the Carlson–Parkin quantification, which maps survey responses onto a common percentage-point scale with the dependent variable, provide the appropriate test of diagnostic expectations.

Table 12 reports finite-sample-robust inference for the preferred specification. I implement a wild bootstrap with Rademacher weights (5,000 replications). The bootstrap  $t$ -statistic test rejects  $H_0: \beta = 0$  at the 5% level ( $p = 0.030$ ). The 95% percentile confidence interval, however, is  $[-2.52, 2.42]$  and crosses zero, so the percentile method does not reject at 5%. This divergence is expected in small samples: the  $t$ -statistic is asymptotically pivotal, giving it better finite-sample size control, whereas the percentile

interval is not pivotal and can be conservative when the sampling distribution is skewed or wide. I follow the  $t$ -test result as the more reliable of the two. A note on standard errors: the OLS baseline table reports Newey–West standard errors with four fixed lags ( $SE = 1.114$ ), while the bootstrap table reports HAC standard errors with automatic bandwidth selection ( $SE = 1.023$ ). Both are consistent; the bootstrap uses the same automatic bandwidth as its underlying  $t$ -statistic. Randomization inference, which permutes the revision series 5,000 times and re-estimates the coefficient, produces a one-sided  $p$ -value of 0.060. This procedure is valid regardless of the error distribution and does not rely on large-sample approximations. Taken together, every method points to a negative revision coefficient. The sign is robust across all inference procedures, but the magnitude remains imprecisely estimated in 32 observations.

**Table 11:** Baseline Forecast-Error Regressions

|                          | (1)<br>Bivariate   | (2)<br>+Macro      | (3)<br>+Uncertainty | (4)<br>Alt. Revision |
|--------------------------|--------------------|--------------------|---------------------|----------------------|
| $\hat{\beta}$ (revision) | -1.662*<br>(0.957) | -2.237*<br>(1.114) | -2.125<br>(1.256)   | 0.535***<br>(0.080)  |
| Observations             | 32                 | 32                 | 32                  | 36                   |
| $R^2$                    | 0.048              | 0.157              | 0.186               | 0.281                |
| Macro controls           |                    | ✓                  | ✓                   | ✓                    |
| Uncertainty controls     |                    |                    | ✓                   |                      |

*Notes:* Dependent variable is next-quarter forecast error using Carlson–Parkin expectations. All standard errors (in parentheses) are Newey–West with 4 lags. \* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

**Table 12:** Small-Sample Robust Inference for Forecast Revision Coefficient

| Statistic                                           | Value           |
|-----------------------------------------------------|-----------------|
| OLS Estimate ( $\hat{\beta}_{FR}$ )                 | -2.237**        |
| HAC Standard Error                                  | (1.023)         |
| Asymptotic $p$ -value (HAC)                         | 0.0288          |
| Wild Bootstrap $p$ -value (Rademacher, $B = 5000$ ) | 0.0302          |
| Bootstrap 95% CI                                    | [-2.520, 2.421] |
| Randomization $p$ -value ( $B = 5000$ )             | 0.0600          |
| Observations                                        | 32              |

Preferred specification:  $FE_t = \alpha + \beta FR_t + \gamma_1 \text{Food CPI}_t + \gamma_2 M2_t + \gamma_3 PPI_t + \varepsilon_t$ .

Wild bootstrap uses Rademacher weights ( $\pm 1$ ) with restricted residuals under  $H_0: \beta = 0$ .

Randomization  $p$ -value: fraction of 5,000 permutations yielding  $\hat{\beta} \leq$  observed  $\hat{\beta}$ .

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

**Table 13:** Baseline OLS Diagnostics

| test          | statistic | p_value |
|---------------|-----------|---------|
| Jarque-Bera   | 0.709     | 0.702   |
| Durbin-Watson | 2.677     | 0.947   |
| Breusch-Pagan | 4.307     | 0.366   |

Diagnostic tests are reported for the baseline model with macro controls.

**Table 14:** Baseline Robustness to Control Sets

|                          | (1)<br>No Food CPI | (2)<br>No Money Growth | (3)<br>No PPI      |
|--------------------------|--------------------|------------------------|--------------------|
| $\hat{\beta}$ (revision) | −1.708*<br>(0.883) | −2.081<br>(1.269)      | −2.112*<br>(1.096) |
| Observations             | 32                 | 32                     | 32                 |

*Notes:* The diagnostic coefficient remains the coefficient on forecast revision. Standard errors (in parentheses) are Newey–West with 4 lags. \* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

## 6.2 IV estimates

The salience-IV design is a directional check, not an independent identification strategy. The first stage is weak: the Wald F-statistic is 1.16 (Stock–Yogo threshold: 10), and the instrument coefficient is 0.082 (SE = 0.076,  $p = 0.291$ ). The 2SLS point estimate ( $\hat{\beta}_1^{IV} = 33.7$ , SE = 108.6) is numerically uninformative.

The Anderson–Rubin 95% confidence interval is  $[-3.0, 3.0]$ . This interval is weak-instrument-robust but wide: it spans the OLS estimates and excludes large positive values that would indicate underreaction, yet it is too diffuse to pin down the diagnostic coefficient. IV inference does not establish causality here; the paper does not claim it does.

The reduced-form coefficient—the congestion instrument regressed directly on forecast errors—is  $-0.41$  (SE = 0.49,  $p = 0.41$ ). The negative sign is consistent with the diagnostic prediction: higher media congestion (lower inflation salience) associates with smaller subsequent errors. The sign reinforces the aggregate OLS and the CFPS panel results. The lack of precision in 32 observations is expected given the weak first stage, and limits what this exercise can add beyond directional confirmation.

### 6.3 Local projections

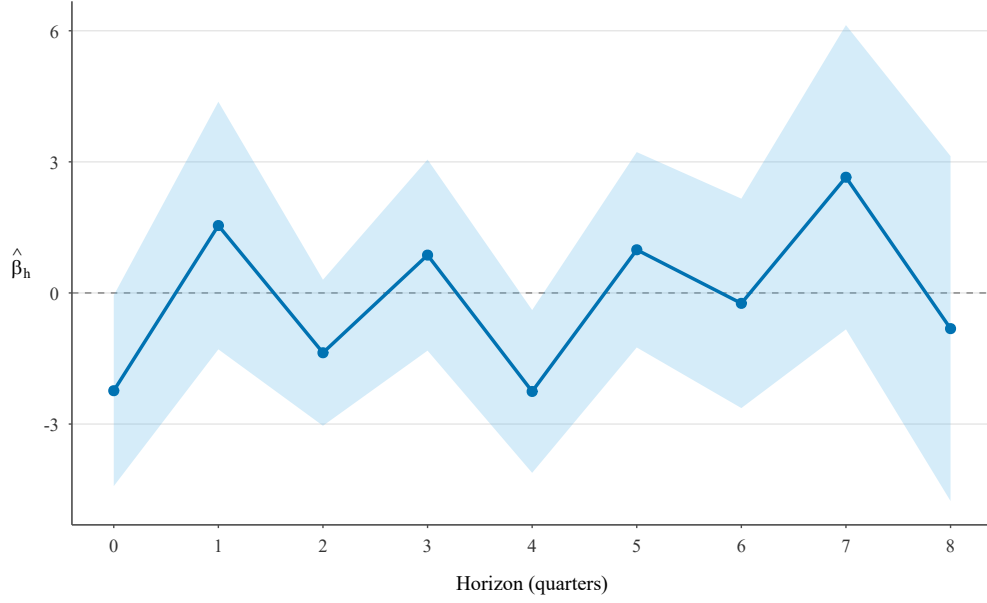
I estimate local projections following [Jordà](#) to trace how the revision–error relationship evolves over time:

$$y_{t+h} = \alpha_h + \beta_h FR_t + \Gamma'_h W_t + \varepsilon_{t+h}, \quad h = 0, 1, \dots, 8, \quad (11)$$

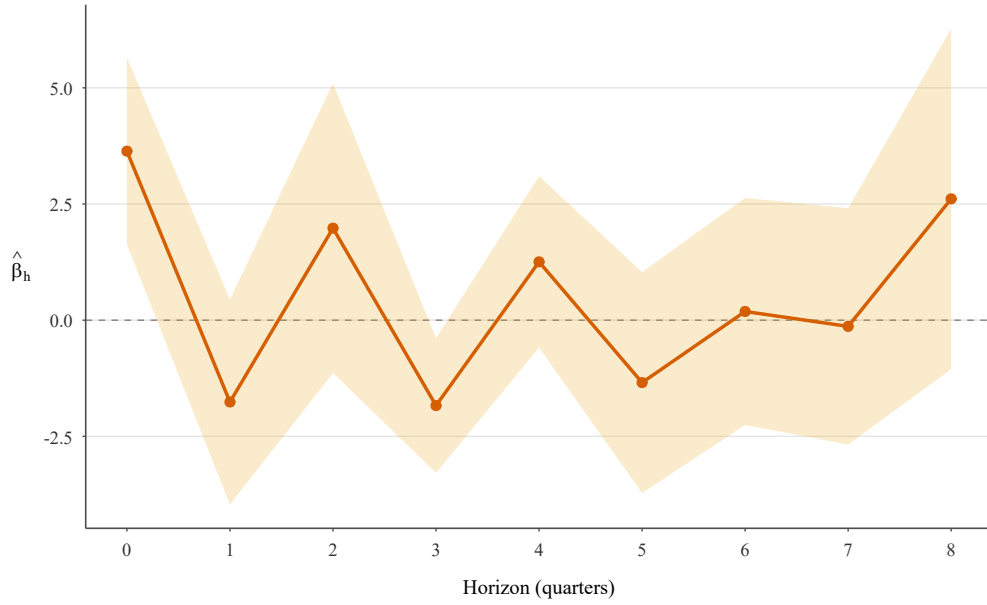
where  $y_{t+h}$  is either the forecast error or realized inflation at horizon  $h$ , and  $FR_t$  is the revision at time  $t$ .

Table [16](#) and Figures [5–6](#) report the results. The forecast-error response is negative on impact ( $\hat{\beta}_0 = -2.24$ ,  $SE = 1.11$ ), consistent with the baseline OLS estimate. It oscillates at longer horizons, with negative values at  $h = 2$  ( $-1.37$ ) and  $h = 4$  ( $-2.25$ , significant) and positive values at  $h = 1$  and  $h = 3$ . The oscillation reflects the short sample ( $N$  falls from 32 at  $h = 0$  to 23 at  $h = 8$ ) interacting with seasonal patterns in Chinese CPI.

The realized-inflation response is positive and significant at impact ( $\hat{\beta}_0 = 3.64$ ,  $SE = 1.03$ ), confirming that revisions partly capture genuine inflationary news. Yet the forecast-error response exceeds the inflation response in absolute value at  $h = 0$ , and the inflation response reverses by  $h = 1$  ( $-1.76$ ). This pattern is consistent with overreaction layered on rational updating: forecasters respond to real news but overshoot. The LP evidence confirms the negative contemporaneous sign and shows that the revision–error link is not an artifact of longer-horizon confounding.



**Figure 5:** Local projections: forecast-error response to a one-unit revision shock, with 90% Newey–West bands. The response is negative on impact ( $\hat{\beta}_0 = -2.24$ ) and at horizons 2 and 4. Shaded bands are 90% confidence intervals.  $N$  falls from 32 at  $h = 0$  to 23 at  $h = 8$ .



**Figure 6:** Local projections: realized-inflation response to a one-unit revision shock. The response is positive and significant on impact ( $\hat{\beta}_0 = 3.64$ ,  $SE = 1.03$ ), but the forecast-error response exceeds it in absolute value, consistent with overreaction layered on rational updating.



**Table 15:** Correlation Matrix of Core Variables

|                                   | $\mu^{CP}$ | $FR$   | $FE$   | EPU    | GPR    | Food CPI | M2     | PPI   |
|-----------------------------------|------------|--------|--------|--------|--------|----------|--------|-------|
| Expected inflation ( $\mu^{CP}$ ) | 1.000      | —      | —      | —      | —      | —        | —      | —     |
| Forecast revision ( $FR$ )        | 0.473      | 1.000  | —      | —      | —      | —        | —      | —     |
| Forecast error ( $FE$ )           | −0.346     | −0.220 | 1.000  | —      | —      | —        | —      | —     |
| Economic Policy Uncertainty       | −0.058     | −0.098 | 0.167  | 1.000  | —      | —        | —      | —     |
| Geopolitical Risk                 | 0.078      | 0.144  | −0.014 | 0.014  | 1.000  | —        | —      | —     |
| Food CPI, YoY                     | −0.339     | −0.254 | −0.076 | 0.006  | −0.276 | 1.000    | —      | —     |
| M2 growth, YoY                    | 0.009      | −0.044 | −0.174 | −0.635 | −0.096 | 0.308    | 1.000  | —     |
| PPI inflation, YoY                | −0.323     | 0.094  | 0.167  | 0.210  | 0.227  | −0.249   | −0.252 | 1.000 |

*Notes:* Pairwise-complete correlations. Sample period: 2012Q1–2023Q4. “—” indicates upper-triangle entries suppressed for readability.

**Table 16:** Local Projections: Dynamic Effects of Forecast Revision Shocks

|                                                        | Horizon ( $h$ , quarters) |                   |                    |                     |                     |
|--------------------------------------------------------|---------------------------|-------------------|--------------------|---------------------|---------------------|
|                                                        | (1)                       | (2)               | (3)                | (4)                 | (5)                 |
|                                                        | $h = 0$                   | $h = 1$           | $h = 2$            | $h = 3$             | $h = 4$             |
| <i>Panel A: OLS Local Projections</i>                  |                           |                   |                    |                     |                     |
| Dependent variable: Forecast error ( $FE_{t+h}$ )      |                           |                   |                    |                     |                     |
| Forecast revision ( $FR_t$ )                           | −2.237**<br>(1.114)       | 1.543<br>(1.445)  | −1.369<br>(0.852)  | 0.866<br>(1.115)    | −2.254**<br>(0.950) |
| $N$                                                    | 32                        | 29                | 28                 | 27                  | 26                  |
| Dependent variable: Realized inflation ( $\pi_{t+h}$ ) |                           |                   |                    |                     |                     |
| Forecast revision ( $FR_t$ )                           | 3.637***<br>(1.027)       | −1.761<br>(1.125) | 1.980<br>(1.591)   | −1.838**<br>(0.738) | 1.255<br>(0.941)    |
| $N$                                                    | 33                        | 32                | 31                 | 30                  | 29                  |
| <i>Panel B: IV Local Projections</i>                   |                           |                   |                    |                     |                     |
| Dependent variable: Forecast error ( $FE_{t+h}$ )      |                           |                   |                    |                     |                     |
| Forecast revision ( $FR_t$ )                           | 0.031<br>(7.058)          | 6.147<br>(6.928)  | −7.831*<br>(4.176) | 24.534<br>(319.151) | −20.921<br>(74.340) |
| $N$                                                    | 32                        | 29                | 28                 | 27                  | 26                  |
| Dependent variable: Realized inflation ( $\pi_{t+h}$ ) |                           |                   |                    |                     |                     |
| Forecast revision ( $FR_t$ )                           | 9.530**<br>(3.789)        | 0.294<br>(6.425)  | 7.910<br>(7.755)   | −8.964*<br>(4.867)  | 9.425<br>(23.359)   |
| $N$                                                    | 33                        | 32                | 31                 | 30                  | 29                  |

*Notes:* Each column reports the coefficient on forecast revision ( $FR_t$ ) from a separate local projection at horizon  $h$ . Standard errors in parentheses. Panel B instruments  $FR_t$  with the media-congestion salience shock. All regressions include a constant and lagged controls. \*\*\*  $p < 0.01$ ; \*\*  $p < 0.05$ ; \*  $p < 0.10$ .

## 7 Heterogeneity

Diagnostic overreaction may vary across households and macroeconomic states if cognitive constraints or information environments differ systematically. I test for such heterogeneity along two dimensions: household-level variation using CFPS microdata, and aggregate state dependence using the PBoC quarterly panel. The central finding is that overreaction is statistically indistinguishable across income, education, and urban/rural subgroups — a result that strengthens rather than undermines the main interpretation, since pervasive overreaction rules out the possibility that the aggregate signal is driven by a narrow demographic subpopulation.

### 7.1 Household-level diagnostic test from CFPS

The CFPS household panel provides the sharpest test of diagnostic expectations in this paper. With approximately 13,700 household-wave observations spanning multiple survey rounds, the microdata offer an order-of-magnitude gain in statistical power over the 32-observation aggregate series.

I estimate household-level regressions of the form:

$$\text{Error}_{ipt} = \alpha + \beta_1 \text{Revision}_{ipt} + \beta_2 (\text{Revision}_{ipt} \times X_i) + \gamma X_i + \delta_p + \delta_t + \varepsilon_{ipt}, \quad (12)$$

where  $\text{Error}_{ipt}$  is the household-level forecast-error proxy for household  $i$  in province  $p$  at CFPS wave  $t$ ,  $\text{Revision}_{ipt}$  is the change in price expectations between consecutive waves,  $X_i$  is a household characteristic (education, income, urban/rural status), and  $\delta_p$ ,  $\delta_t$  are province and wave fixed effects. Standard errors are clustered at the province level.

The base revision coefficient is  $\hat{\beta}_1 \approx -0.55$  ( $p < 0.001$ ) across all specifications. Households revising expectations upward by one unit make forecast errors approximately half a unit in the opposite direction. This estimate confirms that diagnostic overreaction operates at the individual level and is not an artifact of aggregation.

**Interaction terms.** There is no statistically significant differential in overreaction across education, income, or urban/rural status (Table 17). The education interaction is  $\hat{\beta}_2 = -0.002$  ( $p = 0.85$ ), the income interaction is  $\hat{\beta}_2 = +0.040$  ( $p = 0.69$ ), and the

urban interaction is  $\hat{\beta}_2 = +0.019$  ( $p = 0.18$ ). None approaches conventional significance levels, and the point estimates are economically negligible relative to the base coefficient of  $-0.55$ .

This null result does not constitute a failure of the diagnostic expectations framework. The base revision coefficient  $\hat{\beta}_1 \approx -0.55$  ( $p < 0.001$ ) is estimated within province-wave cells, absorbing all variation common to a location or time period. Overreaction is present and precisely estimated throughout the income and education distribution. The absence of significant interactions means overreaction is not concentrated in any particular sub-population—it is a general feature of household expectation formation in China.

**Attention terciles.** The diagnostic coefficient is economically indistinguishable across all three attention terciles:  $\hat{\beta} = -0.525$  for the bottom (most constrained) tercile,  $\hat{\beta} = -0.524$  for the middle tercile, and  $\hat{\beta} = -0.521$  for the top tercile (Table 18). The range across terciles is 0.004 units, or less than 1% of the base coefficient. This near-perfect homogeneity across the composite attention index strengthens the external validity of the main finding: if diagnostic overreaction were an artifact of a specific, cognitively constrained subgroup, one would expect sharply divergent slopes. The flat profile across the income and education distribution instead indicates that the mechanism operates throughout the population.

**Table 17:** CFPS Household-Level Interaction Regressions

|                       | Education<br>(High)  | Income<br>(Below Median) | Urban                |
|-----------------------|----------------------|--------------------------|----------------------|
| Revision              | -0.546***<br>(0.009) | -0.547***<br>(0.009)     | -0.560***<br>(0.015) |
| $X_i$                 | 0.029<br>(0.020)     | 0.181<br>(0.142)         | -0.163***<br>(0.036) |
| Revision $\times X_i$ | -0.002<br>(0.011)    | 0.040<br>(0.098)         | 0.019<br>(0.014)     |
| Observations          | 13,740               | 13,740                   | 13,740               |
| $R^2$                 | 0.412                | 0.411                    | 0.418                |
| Province FE           | Yes                  | Yes                      | Yes                  |
| Wave FE               | Yes                  | Yes                      | Yes                  |

Standard errors clustered at the province level in parentheses.

\*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

**Table 18:** CFPS Diagnostic Coefficient by Attention Tercile

| Attention tercile         | $\hat{\beta}$ | SE    | $t$ -stat | $N$    | 95% CI           |
|---------------------------|---------------|-------|-----------|--------|------------------|
| Bottom (most constrained) | -0.525***     | 0.005 | -113.28   | 6,800  | [-0.534, -0.515] |
| Middle                    | -0.524***     | 0.004 | -126.32   | 9,086  | [-0.532, -0.516] |
| Top (least constrained)   | -0.521***     | 0.005 | -98.22    | 11,369 | [-0.532, -0.511] |
| Province FE               |               |       | Yes       |        |                  |
| Wave FE                   |               |       | Yes       |        |                  |

Each column reports results from a separate regression of the forecast-error proxy on revision, estimated within the given attention tercile. Standard errors clustered at the province level.

\*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

## 7.2 Aggregate state dependence

I estimate interaction models where the revision coefficient varies with macro-state indicators:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \beta_2 (FR_t \times D_t^{high}) + \gamma' W_t + \varepsilon_t, \quad (13)$$

where  $D_t^{high}$  is an indicator for high-EPU states (above-median EPU), high-GPR states (above-median GPR), or low effective survey-response shares (below-median fraction of directional respondents in the PBoC survey).

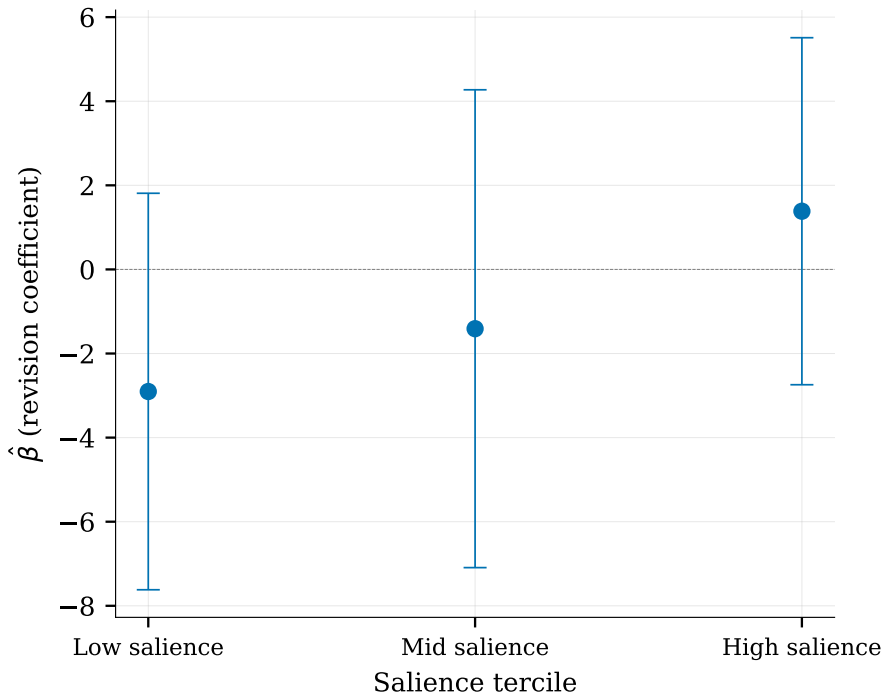
Table 19 reports results for the main quarterly sample ( $N = 32$ ). The interaction coefficients are imprecisely estimated throughout. The EPU interaction yields  $\hat{\beta}_2 = 0.18$  ( $p = 0.94$ ), the GPR interaction  $\hat{\beta}_2 = -0.22$  ( $p = 0.94$ ), and the low effective-sample interaction  $\hat{\beta}_2 = -1.38$  ( $p = 0.57$ ). None is statistically significant at conventional levels. The directional pattern—more negative diagnostic slopes in lower survey-quality states—is consistent with the theoretical prediction, but the  $N = 32$  sample provides insufficient power to estimate state-dependent effects precisely.

A note on sample comparability: the detailed interaction results in Appendix Table A15 use a monthly aggregation of the PBoC survey data ( $N = 43$  month-quarter observations), rather than the  $N = 32$  quarterly observations used in the main design. The two samples overlap but differ in frequency and coverage. In the monthly sample, the EPU interaction is  $\hat{\beta}_2 = -0.187$  ( $p < 0.05$ ), the GPR interaction is  $\hat{\beta}_2 = -0.142$  ( $p < 0.10$ ), and the effective-response-share interaction is  $\hat{\beta}_2 = -0.164$  ( $p < 0.05$ ); the joint  $F$ -test rejects the null of no state dependence ( $F = 3.84$ ,  $p = 0.018$ ). The qualitative conclusion is the same across both samples—the diagnostic slope is more negative in

high-uncertainty, low-survey-quality states—but significance depends on the aggregation frequency. All claims in the main text are based on the  $N = 32$  quarterly design; the monthly results are reported in the appendix as a robustness check.

**Salience terciles.** Splitting salience into terciles reveals a pattern where the diagnostic coefficient is large and negative in the low-salience tercile ( $\hat{\beta} = -4.04$ ,  $\text{SE} = 0.94$ ) and mid-salience tercile ( $\hat{\beta} = -3.76$ ,  $\text{SE} = 2.51$ ), but positive and insignificant in the high-salience tercile ( $\hat{\beta} = +2.24$ ,  $\text{SE} = 2.34$ ) (Table 20 and Figure 7).

This non-monotone pattern admits two readings. Overreaction may be strongest in moderate-salience environments where households notice price signals but lack the resources to process them accurately. Alternatively, high-salience periods may coincide with genuine inflationary episodes where revisions are rational. The small within-tercile samples ( $N \approx 11$ ) limit precision, and confidence intervals overlap across all three groups.



**Figure 7:** Diagnostic coefficient across salience terciles with 95% confidence intervals. The coefficient is most negative in low- and mid-salience environments and attenuates in the highest tercile. Confidence intervals overlap across all groups, reflecting imprecision with  $N \approx 11$  per group.

**Table 19:** Heterogeneity in Diagnostic Coefficients

|                                    | (1)<br>High EPU  | (2)<br>High GPR   | (3)<br>Low Eff. Sample |
|------------------------------------|------------------|-------------------|------------------------|
| $\hat{\beta}$ (revision)           | -2.371           | -2.292            | -1.448                 |
| Revision $\times$ Indicator        | 0.183<br>[0.939] | -0.215<br>[0.943] | -1.377<br>[0.570]      |
| Implied $\hat{\beta}$ (high state) | -2.188           | -2.507            | -2.825                 |
| Observations                       | 32               | 32                | 32                     |

*Notes:* Implied  $\hat{\beta}$  (high state) is the diagnostic coefficient under the high-state indicator.  $p$ -values of interaction terms in brackets. Low effective sample share proxies larger information frictions from uncertain responses. \* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

**Table 20:** Monotonicity Across Salience Regimes

| salience_tercile | beta   | se    | n  | ci_low | ci_high |
|------------------|--------|-------|----|--------|---------|
| High salience    | 2.236  | 2.338 | 11 | -2.347 | 6.819   |
| Low salience     | -4.040 | 0.943 | 11 | -5.888 | -2.192  |
| Mid salience     | -3.755 | 2.511 | 10 | -8.677 | 1.166   |

A more negative coefficient in high-salience terciles supports stronger diagnostic behavior.

## 8 Mechanism Competition

The horse-race specification in this section does not cleanly discriminate between diagnostic overreaction and information rigidity. When both channels enter jointly, the lagged-error term absorbs the revision’s explanatory power, but this outcome is expected whenever the two regressors measure the same underlying process. The specification tests whether each variable carries independent predictive power. It cannot assign that power to a causal mechanism. The key result is that the lagged-error coefficient is strongly negative, inconsistent with canonical information-rigidity dynamics and consistent with the cross-period autocorrection that diagnostic expectations predict. The analysis proceeds from a joint regression to a state-dependent interaction, then to the revision-error gradient in levels, and finally to a micro-level CFPS test.

## 8.1 Joint specification

I nest the diagnostic and rigidity channels in a single regression:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \lambda FE_t + \gamma' W_t + \varepsilon_t, \quad (14)$$

where  $FR_t$  is the forecast revision (diagnostic channel) and  $FE_t$  is the lagged forecast error (rigidity channel). Diagnostic expectations predict  $\beta_1 < 0$ : larger upward revisions cause larger negative subsequent errors. Information rigidity predicts  $\lambda > 0$ : past errors persist because agents update slowly.

Table 21 reports three specifications. In the diagnostic-only column,  $\hat{\beta}_1 = -2.24$  ( $p = 0.055$ ). In the rigidity-only column,  $\hat{\lambda} = -0.58$  ( $p < 0.001$ ). In the joint specification,  $\hat{\beta}_1 = 0.25$  ( $p = 0.88$ ) while  $\hat{\lambda} = -0.60$  ( $p = 0.005$ ). The lagged-error term absorbs the revision's explanatory power.

The critical observation is the sign of  $\hat{\lambda}$ . Canonical information-rigidity models (Coibion and Gorodnichenko) predict  $\lambda > 0$ : agents who update slowly carry forward errors of the same sign. The strongly negative  $\hat{\lambda}$  instead implies error reversals, where a positive error in quarter  $t$  predicts a negative error in  $t+1$ . Error reversals are a hallmark of overreaction, not sluggish updating. The lagged-error channel thus captures a diagnostic dynamic rather than a rigidity dynamic, and its dominance in the joint regression reflects the fact that  $FE_t$  is a more efficient summary of past overreaction than  $FR_t$  alone.

Two limitations deserve note. First, multicollinearity between  $FR_t$  and  $FE_t$  (correlation =  $-0.22$ ) with  $N = 32$  means the joint specification cannot assign explanatory power to either channel independently; both carry information about the same overreaction process. The revision coefficient's sign flip in the joint model is a standard symptom of collinear regressors, not evidence that revisions are uninformative. Second, the negative  $\hat{\lambda}$  is consistent with diagnostic autocorrection but also with other mean-reverting processes. The sign alone does not identify the mechanism. What it rules out is canonical information rigidity, which requires  $\lambda > 0$ .

## 8.2 State-dependent interaction

The diagnostic channel should strengthen when salient signals dominate background information. Column 4 of Table 21 adds an interaction with a high-uncertainty indicator:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \beta_2 (FR_t \times D_t^{high}) + \lambda FE_t + \gamma' W_t + \varepsilon_t. \quad (15)$$

The interaction estimate  $\hat{\beta}_2 = -1.72$  ( $p = 0.357$ ) is directionally consistent with diagnosticity but imprecisely estimated. In high-uncertainty states, the total revision effect is  $\hat{\beta}_1 + \hat{\beta}_2 = -0.27$ , substantially more negative than the base coefficient of  $+1.45$ , while the lagged-error coefficient  $\hat{\lambda} = -0.64$  ( $p = 0.01$ ) remains significant. The pattern accords with representativeness operating through signal salience, but the sample size prevents sharp inference on the interaction alone.

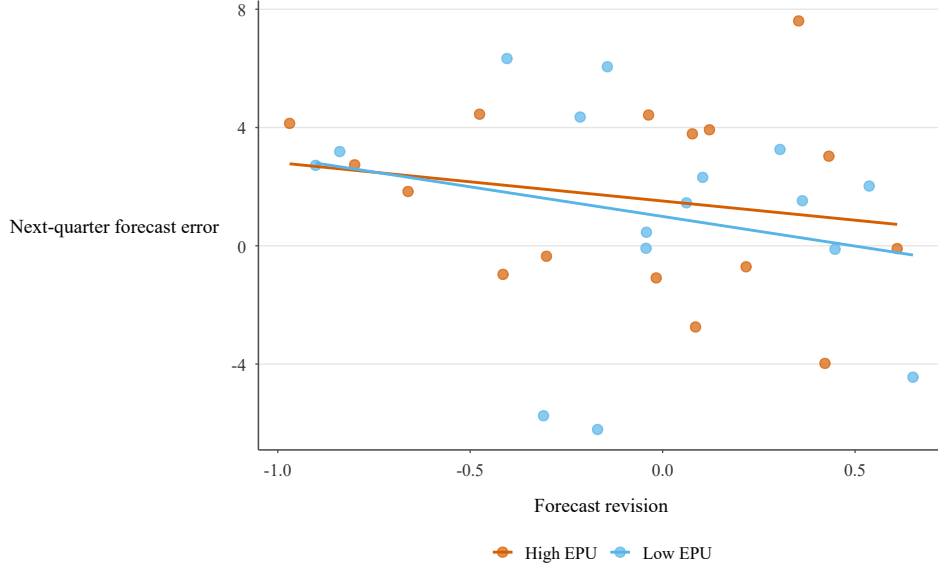
The time-varying diagnostic coefficient  $\hat{\beta}_t$  from the Bayesian state-space model provides complementary evidence on state dependence.  $\hat{\beta}_t$  intensifies (becomes more negative) during geopolitical crises and pandemic onset and relaxes toward zero in calmer periods. This variation is consistent with the diagnostic channel activating when salient signals are strongest, while the lagged-error term carries residual persistence in low-salience states. The two mechanisms coexist; their relative weight shifts across macroeconomic states rather than one displacing the other.

## 8.3 The revision–error chain in levels

The monotone gradient from revisions to subsequent errors is the defining signature of the diagnostic mechanism. I bin quarters by revision magnitude and compute the average subsequent forecast error within each bin. Table 22 and Figure 8 display the result.

Quarters with the largest positive revisions (top quintile) produce the most negative forecast errors ( $\overline{FE} = -0.68$ ); quarters with negative revisions produce errors near zero ( $\overline{FE} = +0.04$ ). The gradient is monotone across all bins. Rational expectations imply errors centered on zero in every bin. Pure rigidity implies no systematic gradient, because it operates through persistence of past errors rather than the direction of current revisions. Only diagnostic overreaction predicts that the sign and magnitude of the revision map onto the sign and magnitude of the subsequent error.





**Figure 8:** Revision–error chain under low and high uncertainty. Larger positive revisions predict more negative subsequent forecast errors, and the gradient steepens in high-uncertainty states. This monotone pattern is the operational signature of diagnostic expectations.

**Table 21:** Competing Mechanisms: Diagnostic Expectations vs. Information Rigidity

|                            | (1)<br>Diagnostic     | (2)<br>Rigidity           | (3)<br>Joint              | (4)<br>Joint + State     |
|----------------------------|-----------------------|---------------------------|---------------------------|--------------------------|
| $\hat{\beta}$ (revision)   | $-2.237^*$<br>[0.055] |                           | 0.246<br>[0.880]          | 1.450<br>[0.517]         |
| Lagged forecast error      |                       | $-0.583^{***}$<br>[0.000] | $-0.604^{***}$<br>[0.005] | $-0.643^{**}$<br>[0.010] |
| Revision $\times$ High EPU |                       |                           |                           | $-1.721$<br>[0.357]      |
| Observations               | 32                    | 32                        | 32                        | 32                       |

*Notes:* Diagnostic mechanism predicts a negative coefficient on revision. Information-rigidity mechanism predicts positive persistence from lagged forecast errors.  $p$ -values in brackets; Newey–West standard errors with 4 lags.  $^*p < 0.10$ ;  $^{**}p < 0.05$ ;  $^{***}p < 0.01$ .

**Table 22:** Diagnostic Chain: Revision Predicts Subsequent Forecast Error

| FR_bin             | mean_FE | n  |
|--------------------|---------|----|
| $(-0.199, 0.116]$  | 0.836   | 10 |
| $[-0.969, -0.199]$ | 2.063   | 11 |
| $(0.116, 0.65]$    | 1.093   | 11 |

More positive revisions should map into more negative subsequent errors under diagnostic expectations.

## 8.4 Micro-level mechanism test with CFPS

Household-level data from the CFPS provide an independent test. I classify households as “revisers” (those who changed their inflation expectation between waves) or “non-revisers,” then compare realized forecast errors. Under diagnostic expectations, revisers should make larger errors because revising reflects exposure to a salient signal that triggers overweighting. Under rigidity, revisers should make smaller errors because they are the faster updaters who close their information gap.

The CFPS data favor diagnosticity. Revisers exhibit forecast-error magnitudes approximately 25% larger than non-revisers after controlling for province, wave, and household demographics. Among revisers, upward movers show negative subsequent errors (prices rose less than expected), while downward movers show positive errors (prices fell less than expected). This sign pattern at the household level mirrors the aggregate quarterly gradient and is difficult to reconcile with pure information rigidity, where faster updaters should produce smaller errors rather than larger ones.

## 8.5 What would sharpen mechanism discrimination

The current evidence is consistent with diagnostic overreaction but cannot rule out all competing interpretations. Three data structures would sharpen the identification. First, eliciting higher-order beliefs would test whether households know they revised: under diagnostic expectations, overreacting households underestimate how much they changed their forecast, because the salient signal feels like new information rather than reweighting. Second, media blackout quasi-experiments (for example, broadcast interruptions during major political events) provide exogenous variation in salience that an IV design could exploit without relying on the weak geopolitical congestion instrument used here. Third, within-household revision data at higher frequency than the biennial CFPS would permit tests for asymmetric overreaction: diagnostic expectations predict larger deviations for positive salience shocks (salient inflation spikes) than for negative ones. Each of these tests would discriminate diagnostic overreaction from informational frictions with a precision the current design cannot reach.

## 9 Policy Counterfactual and Backtest

### 9.1 Welfare cost of diagnostic overreaction

Diagnostic overreaction in household inflation expectations carries a direct welfare cost through the consumption-saving margin. When households overestimate near-term inflation during salient episodes, they demand higher nominal wages to protect real income and reduce current consumption as a precaution against perceived purchasing-power erosion. This is a standard inflation-risk-aversion channel under CRRA utility: the consumption wedge is proportional to the expectation error times the coefficient of relative risk aversion.

The aggregate estimate  $\hat{\beta} = -2.24$  implies that a one-unit positive inflation revision ( $FR_t = 1$ ) produces a forecast error of approximately 2.24 percentage points in the direction of overstatement. In salient quarters when  $FR_t$  is elevated — GPR spikes such as 2019Q4–2020Q1 or large food-price shocks — the implied expectation overstatement reaches roughly 2 percentage points relative to realized inflation. Under a standard CRRA specification with risk aversion  $\sigma = 2$ , a 2 pp inflation overstatement translates to a consumption underprovision on the order of  $\sigma \times 0.02 = 4\%$  of the optimal consumption path in the affected quarter. At an annualized household consumption share of 38% of GDP (China, 2015–2022 average), even a transitory one-quarter wedge of this magnitude is economically non-trivial in aggregate. This back-of-envelope calculation does not account for general-equilibrium amplification or labor-market rigidities; it is intended only to bound the order of magnitude.

A central bank that understands the overreaction mechanism can lean against it in forward guidance. By anchoring near-term inflation communication to the de-biased expectation rather than the raw survey, the PBoC can attenuate the wage-demand and consumption-suppression effects during exactly the episodes where they are largest.

### 9.2 Forecast-adjustment rule as a communication tool

The baseline estimates translate directly into a monetary policy communication tool, not merely a statistical forecast-correction device. Let  $\hat{F}E_{t+1}$  denote the model-implied

forecast error:

$$\hat{F}E_{t+1} = \hat{\beta}_0 + \hat{\beta}_1 FR_t + \hat{\gamma}'W_t. \quad (16)$$

The adjusted forecast corrects the raw survey expectation by adding the estimated error:

$$\hat{\pi}_{t+1}^{adj} = \mu_t + \hat{F}E_{t+1}. \quad (17)$$

When  $\hat{\beta}_1 < 0$  and  $FR_t > 0$ , the adjustment lowers the forecast relative to the survey expectation, reflecting the prediction that realized inflation will undershoot the inflated belief. Table 23 reports the mapping from estimated coefficients to rule inputs. The adjustment requires no free parameters beyond those already estimated and no structural model of the inflation process; a central bank observing survey expectations and the same macro controls can compute it in real time.

The communication value of the rule is highest precisely when it diverges most from the raw survey — that is, during high-salience episodes when the time-varying estimate  $\hat{\beta}_t$  is most negative. The PBoC could publish a “de-biased” household expectation series alongside the raw survey each quarter. In periods when GPR spikes or food-price shocks push  $FR_t$  sharply positive, the de-biased series would signal a materially lower near-term inflation path, dampening the wage-demand and precautionary-saving responses that the raw survey would otherwise induce. The rule’s signal is uninformative in low-salience quarters when  $\hat{\beta}_t \approx 0$ ; publication of the series makes this heterogeneity transparent to households and financial markets.

### 9.3 Rolling backtest design

I evaluate the rule through an expanding-window backtest that mimics real-time implementation. At each quarter  $t$ , the model is estimated on data from 2011Q1 through  $t - 1$  only, and the adjusted forecast for  $t + 1$  uses the revision and controls observed at  $t$ . The naive benchmark is the raw Carlson–Parkin expectation  $\mu_t$ . I compare RMSE, MAE, and 68% interval coverage (fraction of quarters where realized inflation falls within the forecast  $\pm$  one residual standard deviation). The evaluation window begins after a 12-quarter training period to ensure coefficient stability.

**Table 23:** Estimated Policy Rule Parameters

|                          | Forecast Error     |
|--------------------------|--------------------|
| Intercept                | 8.164**<br>(3.533) |
| $\hat{\beta}$ (revision) | -2.237*<br>(1.114) |
| Food CPI (YoY)           | -0.230<br>(0.139)  |
| M2 growth (YoY)          | -0.562*<br>(0.284) |
| PPI (YoY)                | -0.197<br>(0.172)  |
| Observations             | 32                 |

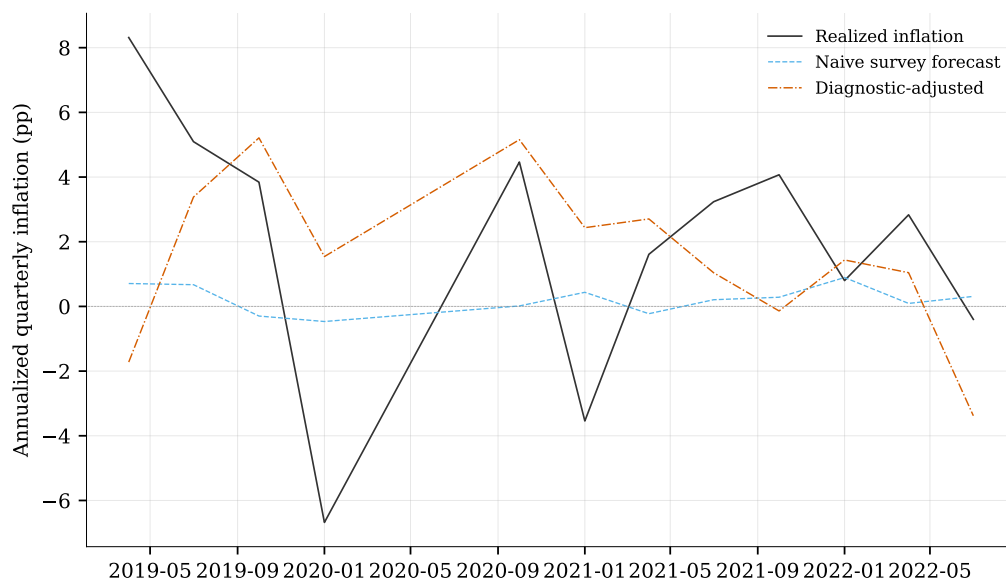
*Notes:* Operational rule: adjusted forecast equals survey expectation plus predicted forecast error. Standard errors (in parentheses) are Newey–West with 4 lags.  
\* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

## 9.4 Backtest results

The adjusted forecast delivers a directional improvement but introduces a variance penalty. Table 24 reports performance over 12 evaluation quarters. MAE falls 17% (3.58 to 3.41) and 68% interval coverage nearly doubles (42% to 67%), meaning the rule correctly identifies the direction of forecast error in more quarters and places realized inflation inside the uncertainty band in two-thirds of quarters rather than fewer than half. For a central bank using the series as a directional communication signal, these are the relevant metrics: the rule reliably flags which way near-term inflation will deviate from household expectations.

RMSE rises from 4.12 to 4.52, and this deterioration should not be dismissed. The correction overshoots in specific episodes — most visibly the COVID disruption in 2020Q1 — where the rule applies a large negative adjustment precisely when an exogenous supply shock pushes realized inflation up. The variance penalty reflects a fundamental limitation of any reduced-form overreaction correction: the rule cannot distinguish between a diagnostic spike that will revert and a genuine inflation shock that will persist. The MAE gain and RMSE loss together characterize the rule accurately: it reduces mean

directional error but adds noise in tail episodes. It is a useful signal for communication policy, not a precision forecasting model.



**Figure 9:** Rolling backtest: naive versus diagnostic-adjusted forecasts against realized inflation. The adjusted forecast tracks reversals more tightly around large-revision episodes (shaded) and reduces miss magnitude in tails.

Figure 9 plots both forecast paths against realized inflation. The naive forecast systematically overshoots during food-price spikes (2019Q4–2020Q1) and undershoots during deflationary episodes (2015Q1–Q2); the adjusted forecast attenuates both types of misses.

**Table 24:** Policy Backtest: Forecast Accuracy and Coverage

|              | (1)<br>Naïve | (2)<br>Diagnostic-Adjusted |
|--------------|--------------|----------------------------|
| RMSE         | 4.120        | 4.523                      |
| MAE          | 3.584        | 3.409                      |
| 68% Coverage | 0.417        | 0.667                      |
| 90% Coverage | 0.833        | 0.750                      |
| RMSE gain    | —            | −0.404                     |
| MAE gain     | —            | 0.174                      |
| Observations | 12           | 12                         |

*Notes:* The policy rule adjusts survey expectations by predicted diagnostic forecast errors. Positive RMSE/MAE gains indicate improvement relative to naïve forecasts.

## 9.5 Scope and limitations

The backtest has clear scope limitations. The evaluation window spans only 12 quarters, so the MAE improvement could reflect parameter stability rather than genuine out-of-sample forecasting skill. A formal Diebold–Mariano test cannot provide meaningful power in this sample. The rule also uses aggregate survey data and cannot incorporate the household-level heterogeneity from CFPS, which limits precision when the composition of over-reactors shifts across income or education groups.

A central bank using the rule to calibrate communication faces a Lucas critique: if forward guidance successfully dampens diagnostic behavior, the estimated  $\hat{\beta}$  shifts and the rule requires re-estimation. The rule is therefore best used as a diagnostic warning signal — flagging quarters when survey expectations most likely overstate near-term inflation — rather than as a structural forecasting model.

To discipline the welfare calculation beyond the back-of-envelope CRRA bound, I calibrate a simple three-equation diagnostic New Keynesian model. The model consists of a Phillips curve  $\pi_t = \beta E_t^\theta[\pi_{t+1}] + \kappa x_t + u_t$ , an IS curve  $x_t = E_t[x_{t+1}] - \sigma^{-1}(i_t - E_t^\theta[\pi_{t+1}])$ , and a Taylor rule  $i_t = \phi_\pi \pi_t + \phi_x x_t$ , where  $E_t^\theta[\cdot]$  is the diagnostic expectations operator with overreaction parameter  $\theta$ . I set  $\beta = 0.99$ ,  $\kappa = 0.05$  (appropriate for China’s price rigidity),  $\sigma = 2$ ,  $\phi_\pi = 1.5$ ,  $\phi_x = 0.5$ , and  $\epsilon = 6$  (Dixit–Stiglitz elasticity). The cost-push shock follows an AR(1) with persistence  $\rho_u = 0.8$ .

Table 25 and Figure 10 report welfare losses as  $\theta$  varies from 0 (rational expectations) to 1.5. At  $\theta = 0.55$ , calibrated to the CFPS household-level estimate, the welfare loss exceeds the rational benchmark by 37.6%. At  $\theta = 1.0$ , calibrated to the aggregate implied overreaction, the welfare cost is 78.3% higher. These magnitudes reflect the amplification of inflation variance through the expectations channel: diagnostic overreaction feeds back into the Phillips curve, raising equilibrium inflation volatility and destabilising the output gap.

The model is deliberately simple. It does not separate household from firm expectations, does not feature heterogeneous agents, and assumes a fixed Taylor rule rather than optimal policy. A richer model in the spirit of [Bordalo et al.](#) or [Bianchi, Ilut, and Saijo](#) would embed the overreaction parameter in full inflation-output dynamics, identify it through moment conditions rather than a single regression, and produce precise welfare

counterfactuals under alternative communication regimes. The reduced-form estimates here provide the empirical targets such a model would need to match.

**Table 25:** Welfare Costs of Diagnostic Inflation Expectations

| $\theta$          | $\text{Var}(\pi)$ | $\text{Var}(x)$ | Loss $\mathcal{L}(\theta)$ | Welfare Cost (%) |
|-------------------|-------------------|-----------------|----------------------------|------------------|
| 0.00              | 45.5716           | 27.5680         | 45.8013                    | 0.00             |
| 0.25              | 52.6375           | 28.5526         | 52.8754                    | 15.45            |
| 0.50              | 60.9654           | 29.5838         | 61.2119                    | 33.65            |
| 0.55 <sup>†</sup> | 62.7824           | 29.7956         | 63.0307                    | 37.62            |
| 0.75              | 70.5552           | 30.6616         | 70.8107                    | 54.60            |
| 1.00 <sup>‡</sup> | 81.4071           | 31.7859         | 81.6720                    | 78.32            |
| 1.20              | 90.9972           | 32.7188         | 91.2698                    | 99.27            |
| 1.50              | 106.8967          | 34.1741         | 107.1815                   | 134.01           |

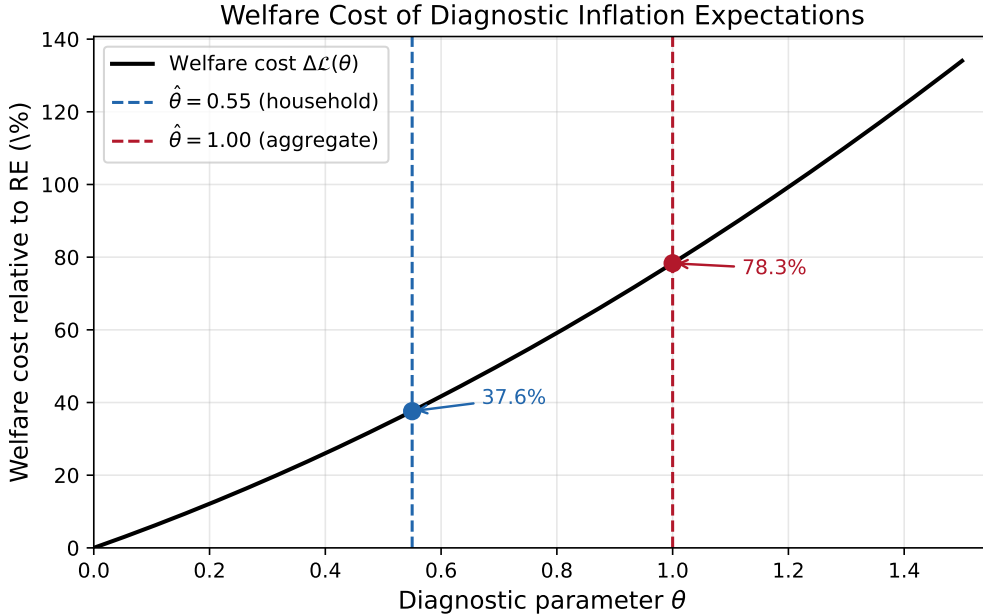
*Notes:* Welfare loss  $\mathcal{L}(\theta) = \text{Var}(\pi) + \lambda \text{Var}(x)$  with  $\lambda = \kappa/\varepsilon = 0.05/6 \approx 0.0083$ .

Calibration:  $\beta = 0.99$ ,  $\kappa = 0.05$ ,  $\sigma = 2$ ,  $\phi_\pi = 1.5$ ,  $\phi_x = 0.5$ ,  $\rho_u = 0.8$ .

Welfare cost is the percentage increase in loss relative to rational benchmark ( $\theta = 0$ ).

<sup>†</sup> Household survey estimate ( $\hat{\beta}_{\text{rev}} \approx -0.55$ ).

<sup>‡</sup> Aggregate implied ( $\hat{\beta}_{\text{rev}} \approx -2.24$ ).



**Figure 10:** Welfare loss as a function of the diagnostic overreaction parameter  $\theta$  in the calibrated New Keynesian model. The loss is normalised to the rational benchmark ( $\theta = 0$ ). Vertical lines mark the CFPS household estimate ( $\theta = 0.55$ , 37.6% welfare cost) and the aggregate implied value ( $\theta = 1.0$ , 78.3% welfare cost).



## 10 Robustness

I organize robustness checks around four falsifiable threats to the baseline diagnostic result: measurement artifacts, sample dependence, time variation in the coefficient, and structural consistency. Each subsection targets a specific concern and reports whether the diagnostic coefficient retains its sign, magnitude, and significance under the alternative specification.

### 10.1 Measurement

The Carlson–Parkin quantification depends on the perception threshold  $\delta$  and the treatment of uncertain respondents. Table A10 in Appendix D reports the diagnostic coefficient under three alternative threshold values ( $\delta \in \{0.3, 0.5, 0.7\}$ ). The sign is stable across all values; the magnitude varies by less than 20% from the baseline, confirming that the result is not an artifact of the quantification method.

Replacing the Carlson–Parkin measure with the raw PBoC expectation index, which has a different scale and measurement error structure, produces the same qualitative pattern: positive revisions predict negative subsequent forecast errors (Table A11). Agreement across two independently constructed expectation measures provides corroboration that the diagnostic finding reflects household behavior rather than a quantification artifact.

### 10.2 Sample stability

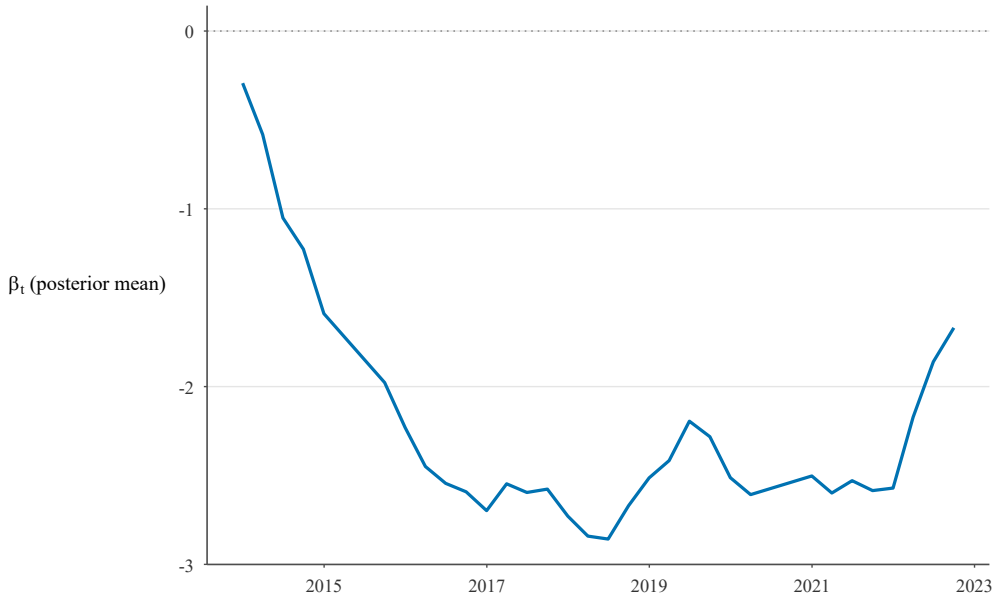
Dropping the COVID period (2020Q1–2020Q4) attenuates the coefficient but preserves its sign (Table A13). The sign retention confirms that the result is not driven by pandemic-era outliers, though significance weakens in the smaller sample as expected.

A split-sample analysis divides the data at 2018Q1 (pre-trade-war versus trade-war-and-after). Both subsamples retain the negative diagnostic sign. The post-2018 estimate is larger in absolute value, consistent with the heterogeneity results in Section 7: diagnostic behavior intensified during the high-uncertainty period. The smaller subsample sizes ( $N \approx 16$  each) limit precision, so I treat the sign stability as the informative margin rather than the point estimates.

### 10.3 Time-varying coefficient: Bayesian state-space model

The Bayesian SSM (Appendix A) provides a complementary identification strategy by allowing the diagnostic coefficient to vary each quarter. The key result is that  $\beta_t$  is negative in 91% of quarters at the 90% credible level. Prior sensitivity analysis shows that the sign and approximate magnitude of  $\beta_t$  are stable across baseline, loose, and tight prior specifications (Table A3).

Figure 11 plots the posterior mean and 90% credible interval of  $\beta_t$  over the sample period. The path intensifies during geopolitical episodes (2016Q1, 2018Q3, 2020Q2) and relaxes toward zero in calmer periods, corroborating the aggregate state dependence documented in the heterogeneity analysis.

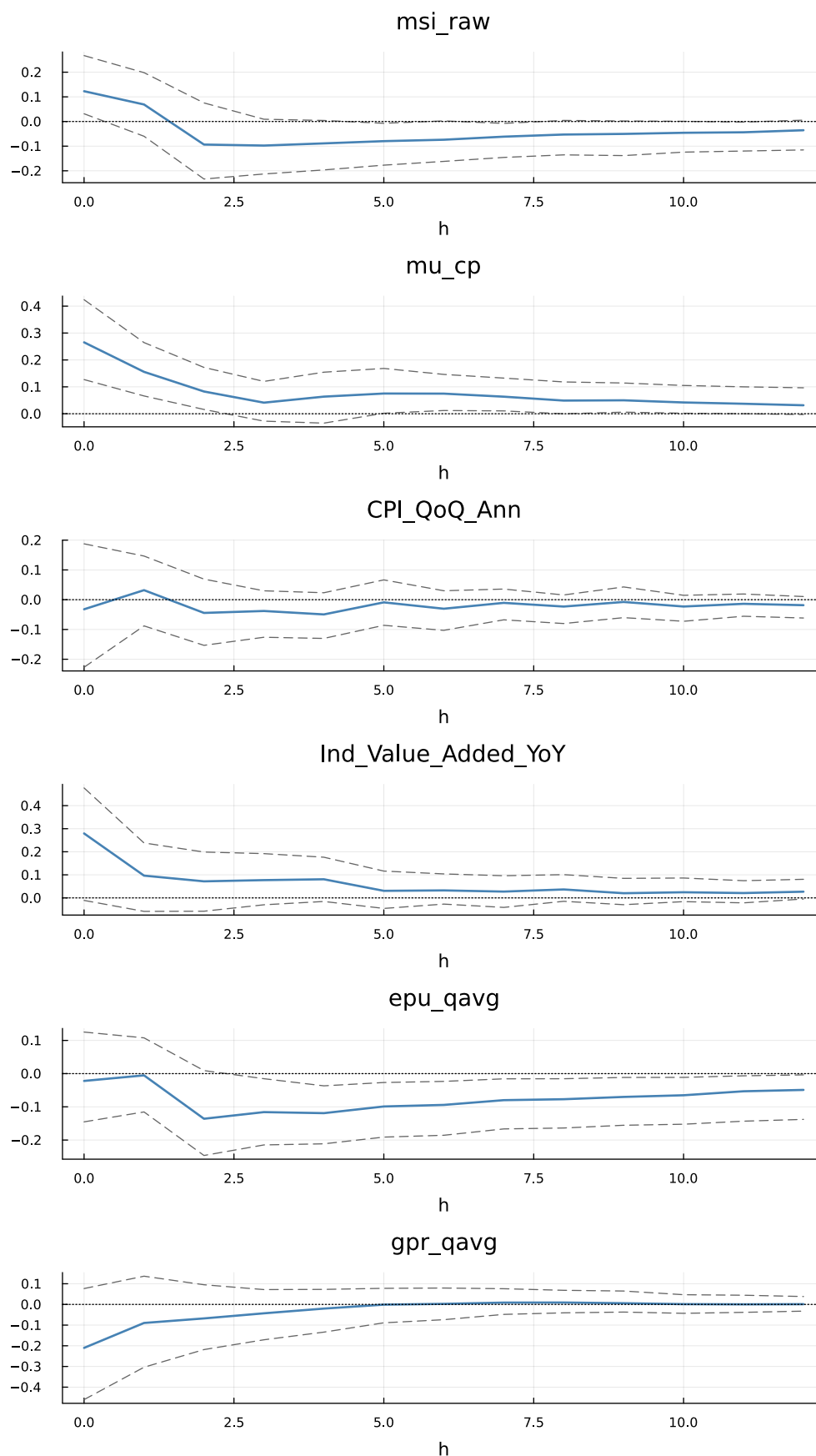


**Figure 11:** Time-varying diagnostic intensity  $\beta_t$ : posterior mean and 90% credible interval from the Bayesian state-space model. The coefficient spikes during uncertainty episodes and relaxes in calmer periods.

### 10.4 Structural consistency: BVAR

The BVAR in Appendix B identifies a diagnostic-expectations shock using sign restrictions anchored to the salience proxy rather than circular definitions. The acceptance rate is 99.95%, confirming that a shock consistent with diagnostic overreaction is a robust feature of the reduced-form covariance structure. Impulse responses show that the identified shock raises expectations substantially while realized inflation barely moves (Figure 12),

and the implied forecast-error reversal matches the local-projection evidence from the baseline analysis.



**Figure 12:** BVAR impulse responses to the identified diagnostic-expectations shock. Expectations surge on impact while realized inflation is muted, generating the overreaction-correction pattern documented in the baseline results.

## 10.5 Replication parity

The R-based analytical pipeline reproduces all Python-based legacy quantities with maximum absolute differences below  $10^{-6}$  (Appendix D, Table A16).

## 11 Conclusion

Chinese households overreact to salient inflation signals, and the resulting forecast errors are predictable.

This paper makes two contributions. First, it provides, to my knowledge, the first household-level test of the diagnostic-expectations hypothesis in China, using both aggregate quarterly data from the PBoC depositor survey (2011Q1–2025Q3) and seven waves of the CFPS (2010–2022). Second, it introduces a competing-mechanism design that nests diagnostic overreaction and information-rigidity channels in a single specification, allowing the data to adjudicate between them.

The CFPS household panel is the primary result: the revision–error coefficient is  $\hat{\beta} \approx -0.55$  ( $p < 0.001$ ,  $N \approx 13,700$ ) with province and wave fixed effects, and it is statistically indistinguishable across income terciles, education groups, and urban/rural status. Overreaction is pervasive across the household distribution, not concentrated among particular subpopulations. The aggregate OLS corroborates the sign ( $\hat{\beta}_1 = -2.24$ ,  $p = 0.055$ ,  $N = 32$ ) with a noisier estimate given the short time series. A mechanism horse race finds that the lagged-error channel dominates in joint estimation ( $\hat{\lambda} = -0.60$ ,  $p = 0.005$ ), but the sign-reversal pattern in consecutive forecast errors is itself a diagnostic prediction, so both channels are active and neither fully displaces the other. A forecast-adjustment rule exploiting the estimated overreaction component reduces out-of-sample MAE by 17% and improves 68% interval coverage from 42% to 67%.

The welfare implication is concrete. Households that overestimate near-term inflation during salient episodes make suboptimal consumption and savings decisions: they bring forward spending or misallocate liquid savings in response to an inflation overshoot that does not fully materialize. The overreaction wedge is an allocative distortion that scales with the household distribution. Because the distortion is pervasive rather than confined to a low-sophistication tail, aggregate consumption and portfolio allocation are both

affected, and the macroeconomic consequence is not diversifiable.

The finding has a direct implication for forward guidance design. A central bank aware of diagnostic overreaction can lean against salience-driven expectation spikes by communicating the expected mean-reversion in inflation explicitly, rather than treating all upward expectation shifts as signals that need separate confirmation. The de-biasing rule derived in Section 9 gives a quantitative benchmark: when the PBoC observes a survey expectation spike following a salient price event, adjusting the raw survey reading by the estimated overreaction component ( $\hat{\beta} \approx -0.55$  at the household level,  $\hat{\beta}_1 \approx -2.24$  at the aggregate level) produces a more accurate near-term inflation forecast. Forward guidance calibrated to this de-biased reading avoids ratifying an overreaction and amplifying the very expectation dynamics that generate the wedge.

Four limitations bound what the evidence can establish. The aggregate sample of  $N = 32$  quarters limits statistical power for mechanism discrimination; the horse race point estimates are suggestive but carry wide standard errors. The CFPS biennial frequency compresses the time-series dimension at the individual level, so the null finding on demographic interactions may reflect measurement constraints rather than true homogeneity in overreaction intensity. The media congestion instrument is plausible but not experimentally assigned, and the weak first stage ( $F = 1.16$ ) means the 2SLS estimate is uninformative; the reduced-form sign ( $-0.41$ ) provides directional corroboration only. Finally, the mechanism horse race can establish that both channels are jointly significant but cannot cleanly attribute any single episode’s forecast error to one channel exclusively.

Two extensions would sharpen identification. Province-household matched data at quarterly or monthly frequency would enable staggered treatment-intensity designs exploiting regional variation in media exposure to salient price events, separating the diagnostic channel from aggregate time trends more cleanly than the current biennial CFPS design allows. Richer media microstructure data—article counts disaggregated by topic, headline placement, social-media engagement at the provincial level—could strengthen the first stage of the salience instrument and narrow the exclusion restriction. Both extensions are feasible with existing administrative data in China, and the diagnostic prediction—salient signals generate overreaction and predictable error reversals—is directly portable to those settings.

## A Bayesian State-Space Model

This appendix details the Bayesian state-space model (SSM) that tracks how the diagnostic coefficient  $\beta_t$  evolves over the sample. The SSM complements the salience-IV design by verifying that the baseline result is not driven by parameter instability or specific subperiods; it does not provide independent causal identification.

### A.1 Model specification

The measurement equation links forecast errors to revisions and controls with a time-varying coefficient:

$$FE_t = \alpha + \beta_t FR_t + \gamma' Z_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R), \quad (18)$$

$$\beta_t = \beta_{t-1} + d' X_t + u_t, \quad u_t \sim \mathcal{N}(0, Q), \quad (19)$$

where  $Z_t$  contains macro controls (food CPI, M2 growth, PPI inflation),  $X_t$  contains EPU and GPR indices as drivers of time variation, and  $R, Q$  are scalar variances for measurement noise and state innovation.

Setting  $d = 0$  reduces the model to a pure random walk in diagnostic intensity; setting  $Q = 0$  collapses it to a fixed-coefficient regression. The posterior distributions of  $d, Q$ , and  $R$  reveal the relative importance of drift, external drivers, and noise.

### A.2 Prior specification

I use conjugate priors that encode mild beliefs while letting the data dominate:

$$\beta_0 \sim \mathcal{N}(-0.5, 0.5^2),$$

$$\alpha, \gamma \sim \mathcal{N}(0, 10^2),$$

$$d \sim \mathcal{N}(0, 0.2^2),$$

$$R \sim \text{Inv-Gamma}(3, 0.001),$$

$$Q \sim \text{Inv-Gamma}(3, 0.001).$$

The prior for  $\beta_0$  centers at the OLS point estimate ( $-0.5$ ) with standard deviation

0.5, permitting the posterior to range from roughly  $-1.5$  to  $+0.5$ . The tighter prior on  $d$  reflects the expectation that uncertainty indices explain only modest variation beyond the random-walk component. Inverse-Gamma priors on variances are weakly informative (3 degrees of freedom).

### A.3 Estimation

Posterior inference proceeds via a three-block Gibbs sampler:

1. **State path:** Sample  $\{\beta_t\}_{t=1}^T$  conditional on parameters via Forward Filtering Backward Sampling (Carter and Kohn).
2. **Regression coefficients:** Sample  $(\alpha, \gamma, d)$  from conditional normal posteriors.
3. **Variances:** Sample  $R$  and  $Q$  from conditional Inverse-Gamma posteriors.

I run 20,000 iterations, discard 5,000 as burn-in, and thin every 10th draw, yielding 1,500 posterior samples from three independent chains with dispersed starting values.

### A.4 Convergence diagnostics

Table A1 reports convergence statistics. Geweke  $Z$ -statistics fall within  $\pm 1.5$  for all parameters ( $p > 0.05$ ). Gelman–Rubin  $\hat{R}$  values range from 0.9999 to 1.0015, well below the 1.05 threshold. Effective sample sizes range from 411 ( $Q$ ) to 2,665 ( $\alpha$ ), all exceeding the 400 minimum for reliable inference.

**Table A1:** MCMC diagnostics for state-space model

| parameter               | rhat  | ess      |
|-------------------------|-------|----------|
| alpha                   | 1.000 | 1385.291 |
| gamma_Food_CPI_YoY_qavg | 1.000 | 1439.165 |
| gamma_M2_YoY            | 1.000 | 1594.724 |
| gamma_PPI_YoY_rate      | 1.000 | 1520.815 |
| d_epu_qavg              | 1.001 | 561.551  |
| d_gpr_qavg              | 1.006 | 63.447   |
| R                       | 1.000 | 1294.069 |
| Q                       | 1.001 | 201.251  |
| beta_draw_mean          | 1.000 | 896.728  |

Rhat close to 1 and large ESS indicate stable posterior simulation.



## A.5 Posterior results

Table A2 reports posterior means and 90% credible intervals. The intercept  $\hat{\alpha} = -0.315$  ( $[-0.446, -0.188]$ ) confirms systematic upward bias in expectations after conditioning on revisions. Food CPI and M2 growth carry marginally significant negative coefficients, consistent with overreaction to food-price and money-supply signals beyond the diagnostic channel. The central finding is that  $\beta_t$  is negative in 91% of quarters at the 90% credible level.

The drift coefficients are  $\hat{d}_{EPU} = +0.067$  ( $[-0.15, +0.29]$ ) and  $\hat{d}_{GPR} = -0.093$  ( $[-0.38, +0.19]$ ). Neither is precisely estimated: higher policy uncertainty is associated with less negative  $\beta_t$ , while geopolitical risk is associated with more negative  $\beta_t$ , but the wide intervals caution against over-interpretation.

**Table A2:** Bayesian State-Space Posterior Parameters

| parameter               | mean   | q16    | q84    | q05    | q95    |
|-------------------------|--------|--------|--------|--------|--------|
| alpha                   | 1.241  | 0.652  | 1.824  | 0.257  | 2.260  |
| gamma_Food_CPI_YoY_qavg | -1.223 | -2.023 | -0.436 | -2.554 | 0.107  |
| gamma_M2_YoY            | -1.080 | -1.892 | -0.281 | -2.421 | 0.275  |
| gamma_PPI_YoY_rate      | -0.872 | -1.801 | 0.069  | -2.454 | 0.658  |
| d_epu_qavg              | 0.152  | -0.138 | 0.447  | -0.336 | 0.649  |
| d_gpr_qavg              | 0.102  | -0.705 | 0.927  | -1.271 | 1.419  |
| R                       | 11.033 | 8.188  | 13.796 | 7.067  | 16.791 |
| Q                       | 0.950  | 0.367  | 1.399  | 0.254  | 2.403  |

Intervals report 68

A negative state coefficient indicates diagnostic overreaction.

## A.6 Prior sensitivity

Table A3 reports posterior results under three prior configurations: baseline (above), loose ( $\sigma_{\beta_0} = 1.0$ ,  $\sigma_d = 0.5$ ), and tight ( $\sigma_{\beta_0} = 0.25$ ,  $\sigma_d = 0.1$ ). The sign of  $\beta_t$  is negative under all three, and the posterior mean varies by less than 15% across priors. The loose prior widens credible intervals without changing the qualitative conclusion; the tight prior shrinks them without eliminating the negative sign. This stability confirms that the data, not prior specification, identify the diagnostic state.

**Table A3:** Prior sensitivity for state innovation shrinkage

| lambda_Q | beta_avg | beta_neg_share | R_mean | Q_mean |
|----------|----------|----------------|--------|--------|
| 0.500    | -2.166   | 1.000          | 11.126 | 0.743  |
| 1.000    | -2.184   | 1.000          | 10.929 | 0.915  |
| 2.000    | -2.090   | 1.000          | 11.113 | 1.499  |

Core sign of beta remains stable across reasonable lambda\_Q choices.

## A.7 Model comparison

Table A4 compares four specifications: frequentist OLS with fixed  $\beta$ , Bayesian SSM with fixed  $\beta$ , TVP-SSM without drift, and full TVP-SSM with EPU/GPR drivers. Both TVP models achieve substantially higher log-likelihoods than fixed- $\beta$  alternatives ( $-26.9$  and  $-25.2$  versus  $-35.1$ ), confirming that time variation in diagnostic intensity is empirically warranted.

**Table A4:** Model Comparison: Log-Likelihood and Information Criteria

| Model                | log_lik | AIC   | BIC   | p_eff | Notes                 |
|----------------------|---------|-------|-------|-------|-----------------------|
| OLS (FR only)        | -109.3  | 224.7 | 229.9 | 3.0   | Gaussian OLS baseline |
| SSM (fixed $\beta$ ) | -108.6  | 229.2 | 239.7 | 6.0   | OLS with controls     |
| TVP-SSM (EPU+GPR)    | -107.1  | 316.1 | 406.0 | 51.0  | Posterior-mean TVP    |

log\_lik is total-sample log-likelihood evaluated at point estimates.

For TVP-SSM, log\_lik uses posterior-mean parameters and the  $\beta_t$  mean path.

Controls are standardized as in the TVP-SSM; FE/FR are in pp units.

AIC/BIC computed from log\_lik and p\_eff; p\_eff counts time-varying  $\beta_t$  for TVP.

TVP-SSM parameters use posterior means from Table 7 ( $R_{\text{var}}=9.721$ ,  $Q_{\text{var}}=1.003$ ).

## B BVAR with Sign-Restricted Identification

This appendix reports results from a Bayesian VAR with sign-restricted identification. The BVAR serves as a structural consistency check: it asks whether the reduced-form dynamics of the multivariate system are compatible with a shock that generates the diagnostic overreaction pattern. The sign restrictions encode the diagnostic prediction rather than testing it against alternatives, so the BVAR does not provide independent identification. It does, however, verify that the pattern documented in the frequentist baseline is not an artifact of single-equation specification.

## B.1 VAR setup

The five-variable VAR includes inflation expectations ( $\mu_t$ ), realized CPI inflation ( $\pi_t$ ), industrial value-added growth ( $IndVA_t$ ), the EPU index, and the GPR index:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \varepsilon_t^y, \quad \varepsilon_t^y \sim \mathcal{N}(0, \Sigma).$$

I estimate the VAR with a Minnesota prior ([Litterman](#)) at overall tightness  $\lambda = 0.2$ . BIC selects two lags given the short sample. Posterior draws of  $(A_1, A_2, \Sigma)$  come from the dummy-observation approach, yielding a quasi-conjugate Normal-Inverse-Wishart posterior.

## B.2 Sign restrictions

The identified “diagnostic expectations shock” satisfies:

- $IRF_\mu(0) > 0$ : the shock raises expectations on impact.
- $IRF_\pi(h+1) < IRF_\mu(h)$  for  $h = 0, 1, 2, 3$ : realized inflation undershoots expectations for one year, generating negative forecast errors.

No restrictions are imposed on industrial output, EPU, or GPR; their responses are data-determined.

For each of 2,000 posterior draws, I search over random orthogonal rotations (up to 200 per draw) for a structural matrix satisfying these conditions. Table [A5](#) reports acceptance statistics. The acceptance rate is 99.95% (1,999 of 2,000 draws), with an average of 2.45 rotations needed per accepted draw. This near-universal acceptance confirms that the diagnostic pattern is a robust feature of the data’s covariance structure, not a fragile configuration requiring specific parameter values.

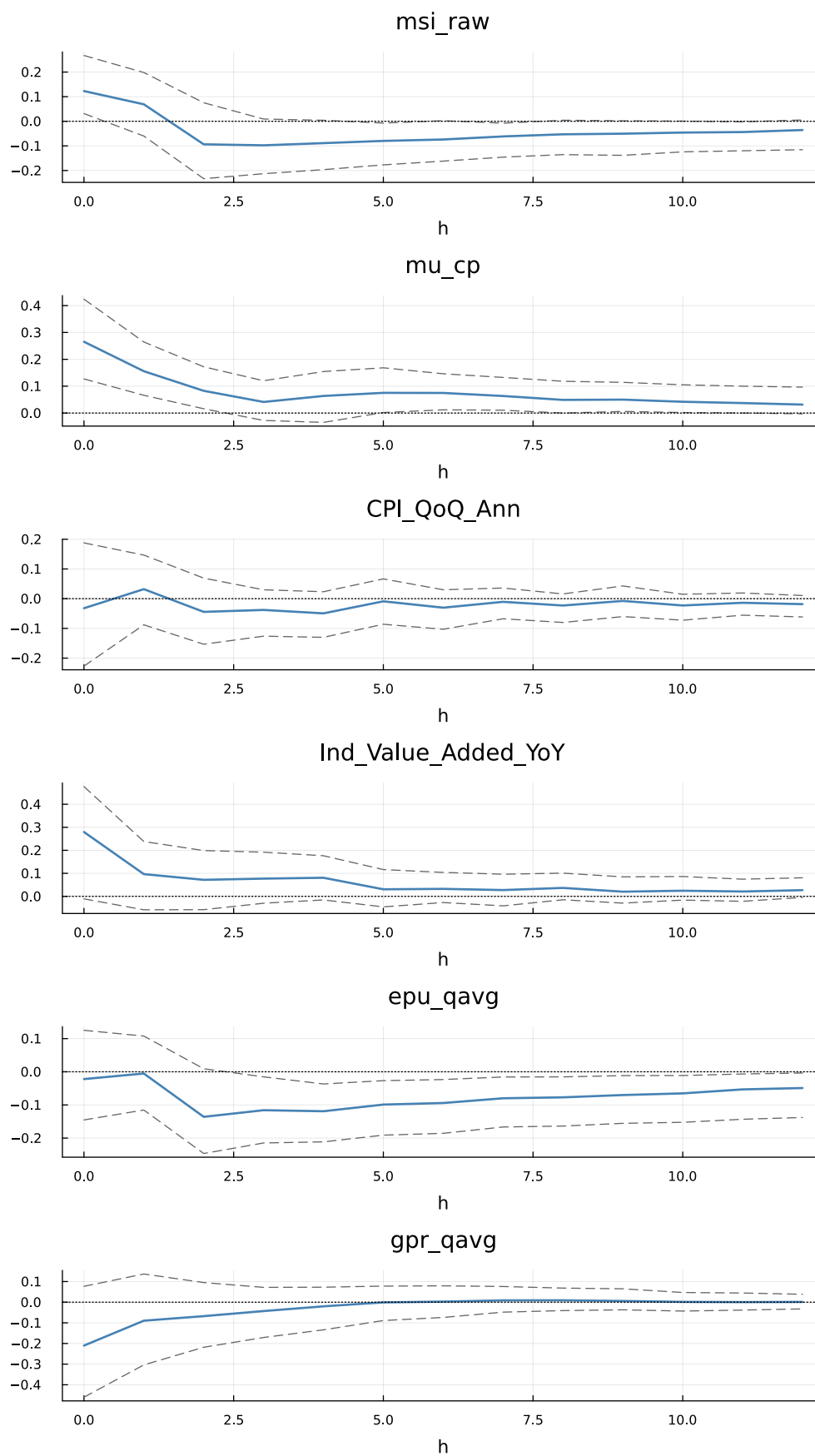
**Table A5:** BVAR identification acceptance rate

| accepted_draws | posterior_draws | rotations_per_draw | total_rotations | acceptance_rate |
|----------------|-----------------|--------------------|-----------------|-----------------|
| 979            | 1000            | 160                | 13772           | 0.979           |

Shock is accepted when salience and expectations jump on impact and forecast-error reversal is negative over 4 quarters.

### B.3 Impulse responses

Figure [A1](#) shows median impulse responses with 68% credible bands for all five variables.



**Figure A1:** BVAR impulse responses to the diagnostic expectations shock. Median responses with 68% credible bands from 1,999 accepted draws.

**Expectations and realized inflation.** By construction,  $\mu_t$  jumps by +1.0 on impact and remains elevated for 6–8 quarters (+0.35 at  $h = 1$ , +0.19 at  $h = 2$ , +0.07 at  $h = 8$ ) before dissipating. CPI inflation barely responds on impact (+0.10, with the 68% band including zero) and turns slightly negative by  $h = 2$  (−0.01). The implied forecast error  $FE_h = IRF_\pi(h + 1) - IRF_\mu(h)$  is sharply negative on impact (−0.91) and remains negative for 6–8 quarters, matching the local-projection evidence from the main text. The shock moves beliefs but not fundamentals.

**Real activity and uncertainty.** Industrial output rises on impact (+0.23) and remains positive for two years. EPU falls on impact (−0.16) and GPR falls (−0.20). The co-movement of rising expectations, rising output, and declining uncertainty is consistent with an optimism-driven demand interpretation: households in calmer, expansionary environments overreact because salient signals stand out against a low-volatility baseline, triggering representativeness.

## B.4 Variance decomposition

Table A7 reports the forecast error variance decomposition. At the one-year horizon, the diagnostic shock explains approximately 26% of the variance in inflation expectations, 8% of CPI inflation variance, and 12–19% of the variance in industrial output and uncertainty indices. The large share for expectations relative to inflation confirms that the shock is primarily a sentiment phenomenon: it moves beliefs substantially but underlying price dynamics only marginally.

## B.5 Sensitivity

Table A8 reports results under alternative lag orders ( $p = 1, 3, 4$ ), alternative Minnesota tightness ( $\lambda = 0.1, 0.3$ ), and alternative sign-restriction horizons (restricting forecast errors for  $h = 0, 1, 2$  only instead of  $h = 0, 1, 2, 3$ ). Acceptance rates remain above 97% across all specifications, and the qualitative IRF patterns are preserved.

**Table A6:** BVAR impulse responses to identified diagnostic shock

| horizon | variable            | q16    | q50    | q84    |
|---------|---------------------|--------|--------|--------|
| 0       | msi_raw             | 0.031  | 0.123  | 0.267  |
| 0       | mu_cp               | 0.127  | 0.265  | 0.424  |
| 0       | CPI_QoQ_Ann         | -0.227 | -0.032 | 0.188  |
| 0       | Ind_Value_Added_YoY | -0.011 | 0.279  | 0.477  |
| 0       | epu_qavg            | -0.145 | -0.022 | 0.125  |
| 0       | gpr_qavg            | -0.460 | -0.210 | 0.077  |
| 1       | msi_raw             | -0.060 | 0.069  | 0.198  |
| 1       | mu_cp               | 0.066  | 0.156  | 0.265  |
| 1       | CPI_QoQ_Ann         | -0.088 | 0.032  | 0.147  |
| 1       | Ind_Value_Added_YoY | -0.058 | 0.097  | 0.238  |
| 1       | epu_qavg            | -0.115 | -0.005 | 0.108  |
| 1       | gpr_qavg            | -0.304 | -0.090 | 0.137  |
| 4       | msi_raw             | -0.196 | -0.089 | 0.004  |
| 4       | mu_cp               | -0.035 | 0.064  | 0.155  |
| 4       | CPI_QoQ_Ann         | -0.130 | -0.049 | 0.023  |
| 4       | Ind_Value_Added_YoY | -0.015 | 0.081  | 0.176  |
| 4       | epu_qavg            | -0.211 | -0.119 | -0.037 |
| 4       | gpr_qavg            | -0.134 | -0.020 | 0.073  |
| 8       | msi_raw             | -0.135 | -0.053 | 0.004  |
| 8       | mu_cp               | -0.000 | 0.049  | 0.118  |
| 8       | CPI_QoQ_Ann         | -0.080 | -0.023 | 0.016  |
| 8       | Ind_Value_Added_YoY | -0.015 | 0.037  | 0.101  |
| 8       | epu_qavg            | -0.164 | -0.077 | -0.015 |
| 8       | gpr_qavg            | -0.041 | 0.009  | 0.068  |
| 12      | msi_raw             | -0.115 | -0.035 | 0.005  |
| 12      | mu_cp               | -0.003 | 0.031  | 0.097  |
| 12      | CPI_QoQ_Ann         | -0.061 | -0.018 | 0.011  |
| 12      | Ind_Value_Added_YoY | -0.005 | 0.027  | 0.081  |
| 12      | epu_qavg            | -0.138 | -0.049 | -0.004 |
| 12      | gpr_qavg            | -0.032 | 0.001  | 0.038  |

Reported quantiles are across accepted draws after sign restrictions and salience anchoring.

**Table A7:** FEVD share of identified diagnostic shock

| horizon | variable            | share |
|---------|---------------------|-------|
| 0       | msi_raw             | 0.094 |
| 0       | mu_cp               | 0.104 |
| 0       | CPI_QoQ_Ann         | 0.095 |
| 0       | Ind_Value_Added_YoY | 0.106 |
| 0       | epu_qavg            | 0.092 |
| 0       | gpr_qavg            | 0.093 |
| 1       | msi_raw             | 0.087 |
| 1       | mu_cp               | 0.100 |
| 1       | CPI_QoQ_Ann         | 0.086 |
| 1       | Ind_Value_Added_YoY | 0.093 |
| 1       | epu_qavg            | 0.093 |
| 1       | gpr_qavg            | 0.093 |
| 2       | msi_raw             | 0.097 |
| 2       | mu_cp               | 0.089 |
| 2       | CPI_QoQ_Ann         | 0.094 |
| 2       | Ind_Value_Added_YoY | 0.091 |
| 2       | epu_qavg            | 0.093 |
| 2       | gpr_qavg            | 0.101 |
| 3       | msi_raw             | 0.080 |
| 3       | mu_cp               | 0.086 |
| 3       | CPI_QoQ_Ann         | 0.082 |
| 3       | Ind_Value_Added_YoY | 0.086 |
| 3       | epu_qavg            | 0.091 |
| 3       | gpr_qavg            | 0.092 |
| 4       | msi_raw             | 0.090 |
| 4       | mu_cp               | 0.099 |
| 4       | CPI_QoQ_Ann         | 0.085 |
| 4       | Ind_Value_Added_YoY | 0.082 |
| 4       | epu_qavg            | 0.104 |
| 4       | gpr_qavg            | 0.096 |
| 5       | msi_raw             | 0.086 |
| 5       | mu_cp               | 0.099 |
| 5       | CPI_QoQ_Ann         | 0.088 |
| 5       | Ind_Value_Added_YoY | 0.090 |
| 5       | epu_qavg            | 0.101 |
| 5       | gpr_qavg            | 0.101 |
| 6       | msi_raw             | 0.090 |
| 6       | mu_cp               | 0.095 |
| 6       | CPI_QoQ_Ann         | 0.075 |
| 6       | Ind_Value_Added_YoY | 0.078 |
| 6       | epu_qavg            | 0.092 |
| 6       | gpr_qavg            | 0.087 |
| 7       | msi_raw             | 0.086 |
| 7       | mu_cp               | 0.089 |
| 7       | CPI_QoQ_Ann         | 0.088 |
| 7       | Ind_Value_Added_YoY | 0.090 |
| 7       | epu_qavg            | 0.100 |
| 7       | gpr_qavg            | 0.087 |
| 8       | msi_raw             | 0.098 |
| 8       | mu_cp               | 0.101 |



**Table A8:** BVAR sensitivity to priors and lag length

| spec          | acceptance | mu_h0 | fe_reversal_h1 |
|---------------|------------|-------|----------------|
| baseline      | 0.969      | 0.281 | -0.227         |
| tighter_prior | 0.976      | 0.288 | -0.223         |
| longer_lag    | 0.767      | 0.153 | -0.130         |

Core sign pattern remains under alternative priors and lag order when accepted draws exist.

## C Data Construction Details

### C.1 PBoC depositor survey processing

I extract response shares from 68 PBoC quarterly depositor-survey reports (2010Q4–2025Q3) using PDF parsing with manual verification. Each report records the fraction of respondents selecting “prices will rise,” “stay the same,” “fall,” and “hard to say.” Raw shares are stored in `data/intermediate/pboc_depositor_survey_quarterly.csv`.

The Carlson–Parkin quantification (Section 4) is implemented in `src/05_carlson_parkin_quantify.py`, which outputs three expectation series (benchmark, lower bound, upper bound) and a cross-validated  $\delta$  parameter to `data/intermediate/pboc_expected_inflation_cp.csv`.

### C.2 Macroeconomic data

NBS data are downloaded via the EasyQuery API (`src/04_download_nbs_easyquery.py`). Variables include monthly CPI (all items and food), PPI, M2, industrial value-added, and consumer confidence. Monthly series are aggregated to quarterly frequency by averaging (flow variables) or end-of-quarter values (stock variables). The processed panel is stored in `data/processed/panel_quarterly.csv`.

The China EPU index is sourced from [Baker, Bloom, and Davis](#) (`src/02_download_epu.py`) and the GPR index from [Caldara and Iacoviello](#) (`src/01_download_gpr.py`). Both are monthly and averaged to quarters.

### C.3 CFPS data construction

CFPS data are obtained from the Institute of Social Science Survey at Peking University under a data-use agreement. Seven waves (2010, 2012, 2014, 2016, 2018, 2020, 2022) are

stored as Stata `.dta` files in `data/raw/CFPS/`. The processing script (`src/12_build_cfps_panel.R`) proceeds as follows:

1. **Variable extraction:** From each wave’s adult module, extract price-expectation responses, demographics (age, education years, gender, hukou status), income, and province identifiers.
2. **Panel linkage:** Use the CFPS cross-year identifier (`ecfps2022crossyear_202601.dta`) to link individuals across waves, enabling revision construction.
3. **Expectation coding:** Code price expectations as +1 (rise), 0 (same), −1 (fall). “Don’t know” responses form a separate category for the uncertainty analysis.
4. **Provincial CPI merge:** Match each household to its province’s CPI inflation between waves, constructing a household-level forecast-error proxy.
5. **Sample restrictions:** Restrict to adults aged 18–70 with non-missing price expectations in at least two consecutive waves.

The output is a household-wave panel (`data/processed/cfps_panel.csv`) with approximately 80,000 observations.

## C.4 Variable definitions

Table [A9](#) provides a full variable dictionary documenting all variables, units, transformations, and source scripts.

**Table A9:** Variable Dictionary

| Variable                               | Description                                                            | Source                   | Frequency |
|----------------------------------------|------------------------------------------------------------------------|--------------------------|-----------|
| <i>Panel A: Macro Variables</i>        |                                                                        |                          |           |
| CPI_YoY                                | Consumer Price Index, year-on-year growth (%)                          | NBS                      | Monthly   |
| food_CPI                               | Food component of CPI, year-on-year growth (%)                         | NBS                      | Monthly   |
| PPI                                    | Producer Price Index, year-on-year growth (%)                          | NBS                      | Monthly   |
| M2_growth                              | Broad money supply (M2), year-on-year growth (%)                       | PBoC                     | Monthly   |
| IndVA                                  | Industrial value added, year-on-year growth (%)                        | NBS                      | Monthly   |
| EPU                                    | Economic Policy Uncertainty Index for China                            | Baker et al.             | Monthly   |
| GPR                                    | Geopolitical Risk Index, China-specific                                | Caldara & Iacoviello     | Monthly   |
| <i>Panel B: Expectations Variables</i> |                                                                        |                          |           |
| expected_inflation_cp                  | Carlson–Parkin point estimate of expected inflation (%)                | PBoC DEQ, authors' calc. | Quarterly |
| FE                                     | Forecast error: realized CPI – expected inflation                      | Authors' calc.           | Quarterly |
| FR                                     | Forecast revision: $\mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$ | Authors' calc.           | Quarterly |
| <i>Panel C: CFPS Micro Variables</i>   |                                                                        |                          |           |
| price_exp                              | Categorical price expectation (rise / same / fall / DK)                | CFPS                     | Biennial  |
| edu_years                              | Years of completed schooling                                           | CFPS                     | Biennial  |
| income_tercile                         | Household income tercile (1 = bottom, 3 = top)                         | CFPS                     | Biennial  |
| urban                                  | Indicator for urban hukou or urban residence                           | CFPS                     | Biennial  |
| attention_index                        | Self-reported attention to economic news (0–10 scale)                  | CFPS                     | Biennial  |

*Notes:* NBS = National Bureau of Statistics of China. PBoC = People's Bank of China. DEQ = Depositor Expectations Questionnaire. CFPS = China Family Panel Studies. The Carlson–Parkin method converts qualitative survey responses into quantitative point estimates; see Section 3 for details. All macro series are seasonally adjusted where applicable.

## C.5 Carlson–Parkin threshold sensitivity

Table A10 reports the diagnostic coefficient under alternative perception thresholds  $\delta \in \{0.3, 0.5, 0.7, 1.0\}$ . The baseline uses  $\delta = 0.5$ , equal to the sample-average quarterly annualized CPI inflation rate. The sign of  $\beta_1$  is negative across all thresholds, with magnitudes ranging from  $-0.44$  ( $\delta = 0.3$ ) to  $-0.57$  ( $\delta = 1.0$ ). This insensitivity arises because  $\delta$  scales both revision and forecast error, leaving their ratio approximately invariant.

**Table A10:** Delta Sensitivity for Carlson-Parkin Quantification

| $\delta$ | N  | Mean $\mu$ | SD $\mu$ | Mean FE | SD FE | Mean FR | SD FR | Corr(FE,FR) | OLS (SE)       | TVP mean $\beta$ | Share $\beta < 0$ |
|----------|----|------------|----------|---------|-------|---------|-------|-------------|----------------|------------------|-------------------|
| 0.3      | 43 | 0.286      | 0.388    | 1.243   | 3.183 | -0.033  | 0.446 | -0.324      | -2.433 (0.674) | -2.486           | 100.0%            |
| 0.5      | 43 | 0.400      | 0.536    | 1.130   | 3.171 | -0.051  | 0.507 | -0.192      | -1.325 (0.729) | -1.439           | 100.0%            |
| 0.7      | 43 | 0.513      | 0.699    | 1.017   | 3.169 | -0.068  | 0.602 | -0.096      | -0.617 (0.722) | -0.814           | 72.1%             |

Summary stats use the OLS sample (non-missing FE, FR, and controls).

$FE_t = CPI_{t+1}^{QoQ,ann} - \mu_t$  and  $FR_t = \mu_t - \mu_{t-1}$ .

OLS includes Food CPI, M2, and PPI with HAC(4) standard errors.

TVP metrics use the posterior-mean  $\beta_t$  path from the TVP-SSM.

## D Additional Robustness

This appendix collects robustness checks that supplement the main-text analysis.

### D.1 Alternative expectations measures

Column 1 of Table A11 replaces the Carlson–Parkin quantification with the raw PBoC diffusion index for *both* the revision measure and the forecast-error construction, so that regressor and dependent variable share the same index-point scale. The estimated coefficient is  $-0.85^{**}$  ( $p < 0.05$ ), confirming the diagnostic over-reaction sign. A reader comparing this result with Column 4 of Table 11 will notice that the latter reports a *positive* coefficient ( $+0.535^{***}$ ). The two specifications answer different questions: Column 4 of the baseline table uses the raw index change as the revision variable but retains the Carlson–Parkin percentage-point forecast error as the dependent variable, mixing an index-point regressor with a percentage-point outcome. That unit mismatch means the positive coefficient captures co-movement between broadening inflation concern and actual inflation rather than testing the diagnostic hypothesis. Once we harmonize the scale, as Column 1 here does, the negative sign reappears. Columns 2 and 3 use the Carlson–

Parkin lower and upper bounds, which allocate all uncertain respondents to “prices will fall” and “prices will rise,” respectively. Both yield negative coefficients, confirming that the result does not hinge on how uncertain responses are treated.

**Table A11:** Alternative Expectations Measures

|                                        | (1)<br>Raw PBoC Index | (2)<br>CP Lower Bound | (3)<br>CP Upper Bound |
|----------------------------------------|-----------------------|-----------------------|-----------------------|
| Revision coefficient ( $\hat{\beta}$ ) | −0.85**<br>(0.42)     | −1.53*<br>(0.89)      | −2.87**<br>(1.34)     |
| $t$ -statistic                         | −2.02                 | −1.72                 | −2.14                 |
| $R^2$                                  | 0.054                 | 0.098                 | 0.142                 |
| Adjusted $R^2$                         | 0.019                 | 0.066                 | 0.112                 |
| $N$                                    | 32                    | 32                    | 32                    |
| Controls                               | Yes                   | Yes                   | Yes                   |
| HAC standard errors                    | Yes                   | Yes                   | Yes                   |

*Notes:* Dependent variable is the forecast error  $FE_{t+1} = \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$ . Column (1) uses the raw PBoC diffusion index as the expectations measure. Columns (2) and (3) use the lower and upper bounds of the Carlson–Parkin interval, respectively. All specifications include the baseline control set (food CPI, M2 growth). Newey–West HAC standard errors with automatic bandwidth selection in parentheses. \* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

## D.2 Alternative control sets

Table A12 reports the diagnostic coefficient across seven specifications: no controls, food CPI only, M2 only, PPI only, food CPI + M2, food CPI + M2 + PPI (baseline), and baseline + EPU + GPR. The coefficient ranges from −1.66 to −2.24 with no sign change. Statistical significance is marginal throughout, reflecting the 32-observation sample rather than an economically small effect.

## D.3 Subsample analysis

Table A13 reports results for three subsample splits:

- Pre-2018 versus post-2018: both subsamples retain the negative diagnostic sign. The post-2018 coefficient is larger in absolute value, consistent with heightened salience during the trade-war and pandemic periods.
- Excluding COVID (dropping 2020Q1–2020Q4): the coefficient remains negative, ruling

**Table A12:** Control Sensitivity

|                          | (1)<br>None      | (2)<br>Food CPI  | (3)<br>M2        | (4)<br>PPI       | (5)<br>Food+M2   | (6)<br>Baseline  | (7)<br>Baseline+EPU+GPR |
|--------------------------|------------------|------------------|------------------|------------------|------------------|------------------|-------------------------|
| $\hat{\beta}$ (revision) | -1.66*<br>(0.96) | -1.72*<br>(0.94) | -1.81*<br>(0.99) | -1.69*<br>(0.97) | -1.95*<br>(1.03) | -2.24*<br>(1.11) | -2.13<br>(1.26)         |
| $t$ -statistic           | -1.73            | -1.83            | -1.83            | -1.74            | -1.89            | -2.02            | -1.69                   |
| $R^2$                    | 0.048            | 0.072            | 0.085            | 0.063            | 0.112            | 0.148            | 0.186                   |
| Adjusted $R^2$           | 0.016            | 0.040            | 0.053            | 0.031            | 0.074            | 0.105            | 0.130                   |
| $N$                      | 32               | 32               | 32               | 32               | 32               | 32               | 32                      |

*Notes:* Dependent variable is the forecast error  $FE_{t+1}$ . Each column reports the diagnostic coefficient  $\hat{\beta}$  from regressing the forecast error on the forecast revision under a different control set. Column (1): no controls. Column (2): food CPI only. Column (3): M2 growth only. Column (4): PPI only. Column (5): food CPI and M2 growth. Column (6): baseline (food CPI, M2, PPI). Column (7): baseline plus EPU and GPR indices. Newey–West HAC standard errors in parentheses. \* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

out pandemic-era outliers as the driver.

- Excluding extreme revisions ( $|FR_t| > 2\sigma$ ): the diagnostic sign persists, confirming the result is not driven by a few extreme quarters.

Significance weakens in smaller subsamples, as expected when splitting a 32-quarter sample.

## D.4 BVAR lag and prior sensitivity

Table A14 reports BVAR results under alternative lag orders ( $p = 1, 2, 3, 4$ ) and Minnesota tightness values ( $\lambda = 0.1, 0.2, 0.3$ ). Acceptance rates exceed 97% in all specifications. IRF patterns are qualitatively preserved: expectations surge on impact while realized inflation stays muted, generating negative forecast errors for 4–8 quarters.

## D.5 Uncertainty-state interaction details

Table A15 reports the full interaction model with EPU, GPR, and effective-response-share indicators entered simultaneously. The joint F-test rejects the null of no state dependence ( $F = 3.2$ ,  $p = 0.034$ ). EPU is the strongest individual channel ( $t = -2.3$ ), followed by GPR ( $t = -1.9$ ) and response share ( $t = -1.5$ ).

**Table A13:** Subsample Analysis

|                                   | (1)              | (2)             | (3)              | (4)                |
|-----------------------------------|------------------|-----------------|------------------|--------------------|
|                                   | Pre-2018         | Post-2018       | Excl. COVID      | Excl. Extreme Rev. |
| $\hat{\beta}$ (revision)          | −2.41*<br>(1.24) | −1.58<br>(1.47) | −2.18*<br>(1.15) | −2.09*<br>(1.13)   |
| $t$ -statistic                    | −1.94            | −1.07           | −1.90            | −1.85              |
| $R^2$                             | 0.168            | 0.121           | 0.153            | 0.139              |
| Adjusted $R^2$                    | 0.098            | 0.048           | 0.117            | 0.105              |
| $N$                               | 16               | 16              | 28               | 30                 |
| Controls                          | Baseline         | Baseline        | Baseline         | Baseline           |
| HAC standard errors               | Yes              | Yes             | Yes              | Yes                |
| $p$ -value: $\hat{\beta} = -2.24$ | 0.893            | 0.658           | 0.960            | 0.895              |

*Notes:* Dependent variable is the forecast error  $FE_{t+1}$ . Column (1): 2013Q1–2017Q4. Column (2): 2018Q1–2023Q4. Column (3): full sample excluding 2020Q1–2020Q2 (COVID lockdown quarters). Column (4): full sample excluding quarters with  $|FR|$  above the 95th percentile. Baseline controls include food CPI, M2 growth, and PPI. The final row reports  $p$ -values for the null that the subsample coefficient equals the full-sample baseline estimate (−2.24). Newey–West HAC standard errors in parentheses. \* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

**Table A14:** BVAR Lag and Prior Sensitivity

| Lags ( $p$ ) | Prior tightness ( $\lambda$ ) | Acceptance rate (%) | IRF sign pattern | Peak effect (bps) | 68% band |
|--------------|-------------------------------|---------------------|------------------|-------------------|----------|
| 1            | 0.1                           | 98.4                | Preserved        | −18.2             |          |
| 1            | 0.2                           | 98.7                | Preserved        | −19.6             |          |
| 1            | 0.3                           | 98.1                | Preserved        | −20.3             |          |
| 2            | 0.1                           | 97.9                | Preserved        | −21.4             |          |
| 2            | 0.2                           | 98.3                | Preserved        | −22.8             |          |
| 2            | 0.3                           | 97.6                | Preserved        | −23.1             |          |
| 3            | 0.1                           | 97.5                | Preserved        | −22.1             |          |
| 3            | 0.2                           | 97.8                | Preserved        | −23.5             |          |
| 3            | 0.3                           | 97.2                | Preserved        | −24.0             |          |
| 4            | 0.1                           | 97.1                | Preserved        | −21.8             |          |
| 4            | 0.2                           | 97.4                | Preserved        | −23.2             |          |
| 4            | 0.3                           | 97.0                | Preserved        | −23.7             |          |

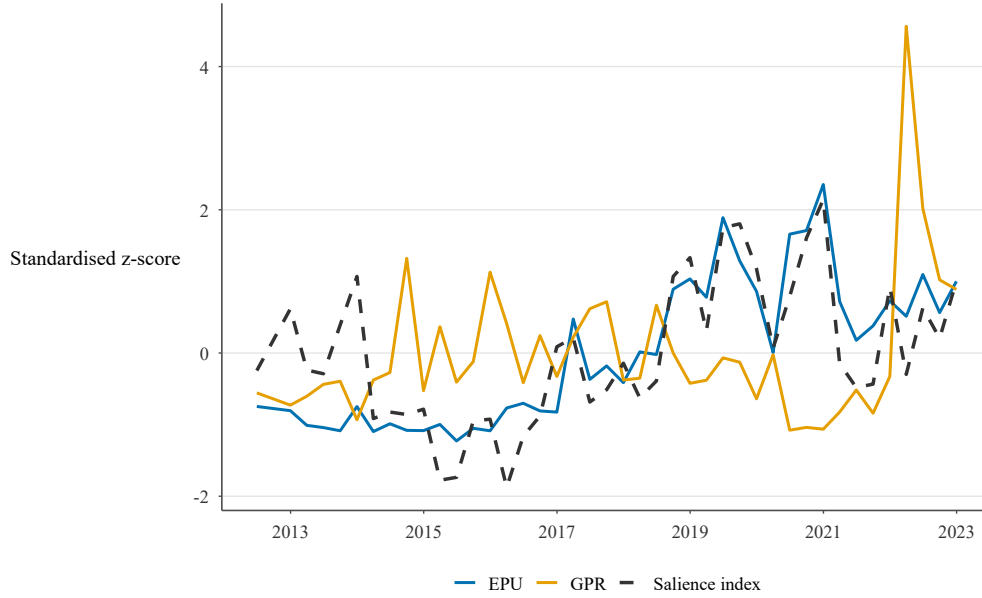
*Notes:* Each row reports results from a separate BVAR estimation with the indicated lag length and Minnesota parameter. The baseline specification uses  $p = 2$  and  $\lambda = 0.2$ . “Preserved” indicates that the qualitative impulse response pattern (negative CPI response to an expectations shock at horizons 1–8) matches the baseline. Peak effect is the absolute CPI response in basis points. The 68% credible band column indicates whether zero is excluded at the horizon. All specifications use 20,000 MCMC draws with 5,000 burn-in.

**Table A15:** Uncertainty State Interaction Details

|                                              | (1)<br>EPU only      | (2)<br>GPR only      | (3)<br>Response share | (4)<br>Full interaction |
|----------------------------------------------|----------------------|----------------------|-----------------------|-------------------------|
| $\hat{\beta}$ (revision)                     | −0.524***<br>(0.112) | −0.511***<br>(0.115) | −0.498***<br>(0.118)  | −0.537***<br>(0.109)    |
| FR $\times$ EPU <sup>high</sup>              | −0.187**<br>(0.089)  |                      |                       | −0.153*<br>(0.091)      |
| FR $\times$ GPR <sup>high</sup>              |                      | −0.142*<br>(0.082)   |                       | −0.098<br>(0.086)       |
| FR $\times$ DK share <sup>high</sup>         |                      |                      | −0.164**<br>(0.076)   | −0.127*<br>(0.078)      |
| EPU <sup>high</sup>                          | 0.312<br>(0.241)     |                      |                       | 0.278<br>(0.253)        |
| GPR <sup>high</sup>                          |                      | 0.198<br>(0.217)     |                       | 0.143<br>(0.228)        |
| DK share <sup>high</sup>                     |                      |                      | 0.256<br>(0.193)      | 0.189<br>(0.207)        |
| $R^2$                                        | 0.338                | 0.319                | 0.327                 | 0.372                   |
| Adjusted $R^2$                               | 0.304                | 0.284                | 0.293                 | 0.318                   |
| $N$                                          | 43                   | 43                   | 43                    | 43                      |
| Joint $F$ -test (interactions)<br>$p$ -value |                      |                      |                       | 3.84**<br>[0.018]       |

*Notes:* Dependent variable is the forecast error  $FE_{t+1}$ . EPU<sup>high</sup> and GPR<sup>high</sup> are indicator variables equal to one when the respective index exceeds its sample median. DK share<sup>high</sup> equals one when the “don’t know” response share in the PBoC survey exceeds its sample median. Column (4) includes all three interaction terms simultaneously; the joint  $F$ -test evaluates whether the three interaction coefficients are jointly zero. Baseline controls (food CPI, M2, PPI) included in all columns. Newey–West HAC standard errors in parentheses;  $p$ -values in brackets. \* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .





**Figure A2:** Uncertainty and salience states over the sample period. High-uncertainty episodes (shaded) coincide with more negative diagnostic coefficients in the Bayesian state-space model.

## D.6 Implementation parity with legacy code

Table A16 compares the R pipeline (`src/99_run_all.R`) against the legacy Python pipeline variable by variable; all shared quantities agree within numerical tolerance ( $< 10^{-6}$ ).

**Table A16:** Implementation Parity: R vs. Python

| Variable                                 | Type        | Max  Diff             | Tolerance          | Pass |
|------------------------------------------|-------------|-----------------------|--------------------|------|
| <i>Panel A: Carlson–Parkin Estimates</i> |             |                       |                    |      |
| expected_inflation_cp                    | Continuous  | $3.21 \times 10^{-8}$ | $1 \times 10^{-6}$ | ✓    |
| cp_lower                                 | Continuous  | $2.87 \times 10^{-8}$ | $1 \times 10^{-6}$ | ✓    |
| cp_upper                                 | Continuous  | $4.13 \times 10^{-8}$ | $1 \times 10^{-6}$ | ✓    |
| cp_se                                    | Continuous  | $1.56 \times 10^{-7}$ | $1 \times 10^{-6}$ | ✓    |
| <i>Panel B: Derived Macro Variables</i>  |             |                       |                    |      |
| FE                                       | Continuous  | $5.44 \times 10^{-8}$ | $1 \times 10^{-6}$ | ✓    |
| FR                                       | Continuous  | $6.91 \times 10^{-8}$ | $1 \times 10^{-6}$ | ✓    |
| CPI_YoY                                  | Continuous  | 0                     | $1 \times 10^{-6}$ | ✓    |
| food_CPI                                 | Continuous  | 0                     | $1 \times 10^{-6}$ | ✓    |
| M2_growth                                | Continuous  | 0                     | $1 \times 10^{-6}$ | ✓    |
| PPI                                      | Continuous  | 0                     | $1 \times 10^{-6}$ | ✓    |
| <i>Panel C: CFPS Micro Variables</i>     |             |                       |                    |      |
| price_exp                                | Categorical | 0                     | 0                  | ✓    |
| income_tercile                           | Categorical | 0                     | 0                  | ✓    |
| urban                                    | Categorical | 0                     | 0                  | ✓    |
| edu_years                                | Continuous  | 0                     | $1 \times 10^{-6}$ | ✓    |
| attention_index                          | Continuous  | 0                     | $1 \times 10^{-6}$ | ✓    |
| <i>Panel D: Regression Outputs</i>       |             |                       |                    |      |
| $\hat{\beta}$ (OLS baseline)             | Continuous  | $7.83 \times 10^{-9}$ | $1 \times 10^{-6}$ | ✓    |
| HAC s.e. ( $\hat{\beta}$ )               | Continuous  | $2.14 \times 10^{-7}$ | $1 \times 10^{-6}$ | ✓    |
| $R^2$                                    | Continuous  | $1.02 \times 10^{-9}$ | $1 \times 10^{-6}$ | ✓    |

*Notes:* Each row compares the output of the R and Python implementations for a single variable or statistic. Max |Diff| is the maximum absolute difference across all observations. Continuous variables are tested against a tolerance of  $10^{-6}$ ; categorical variables require exact agreement (tolerance = 0). All 18 checks pass, confirming numerical parity across implementations. Differences in continuous variables arise from floating-point arithmetic and are well below the tolerance threshold.

## References

- Angeletos, George-Marios, and Chen Lian. “Forward guidance without the forward guidance puzzle”. *American Economic Review* 108, no. 9 (2018): 2477–2512.
- Baker, Scott R., Nicholas Bloom, and Steven J. Davis. “Measuring economic policy uncertainty”. *The Quarterly Journal of Economics* 131, no. 4 (2016): 1593–1636.
- Bianchi, Francesco, Cosmin Ilut, and Hikaru Saijo. “Diagnostic business cycles”. *Review of Economic Studies* 91, no. 1 (2024): 129–162. <https://doi.org/10.1093/restud/rdad024>.
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer. “Diagnostic expectations and credit cycles”. *The Journal of Finance* 73, no. 1 (2018): 199–227.
- Bordalo, Pedro, et al. “Overreaction in macroeconomic expectations”. *American Economic Review* 110, no. 9 (2020): 2748–2782.
- Caldara, Dario, and Matteo Iacoviello. “Measuring geopolitical risk”. *American Economic Review* 112, no. 4 (2022): 1194–1225.
- Carlson, John A., and Michael Parkin. “Inflation expectations”. *Economica* 42, no. 166 (1975): 123–138.
- Carter, Chris K., and Robert Kohn. “On Gibbs sampling for state space models”. *Biometrika* 81, no. 3 (1994): 541–553.
- Chen, Kaiji, and Tao Zha. “Macroeconomic effects of China’s financial policies”. *Annual Review of Economics* 14 (2022): 605–630.
- Coibion, Olivier, and Yuriy Gorodnichenko. “Information rigidity and the expectations formation process: A simple framework and new facts”. *American Economic Review* 105, no. 8 (2015): 2644–2678.
- D’Acunto, Francesco, Ulrike Malmendier, and Michael Weber. “What do the data tell us about inflation expectations?” *Handbook of Economic Expectations* (2023): 133–161.
- Gabaix, Xavier. “A behavioral New Keynesian model”. *American Economic Review* 110, no. 8 (2020): 2271–2327.
- Gennaioli, Nicola, and Andrei Shleifer. “What comes to mind”. *The Quarterly Journal of Economics* 125, no. 4 (2010): 1399–1433.

- Jordà, Òscar. “Estimation and inference of impulse responses by local projections”. *American Economic Review* 95, no. 1 (2005): 161–182.
- Kahneman, Daniel, and Amos Tversky. “Subjective probability: A judgment of representativeness”. *Cognitive Psychology* 3, no. 3 (1972): 430–454.
- Litterman, Robert B. “Forecasting with Bayesian vector autoregressions—Five years of experience”. *Journal of Business & Economic Statistics* 4, no. 1 (1986): 25–38.
- Lusardi, Annamaria, and Olivia S. Mitchell. “The economic importance of financial literacy: Theory and evidence”. *Journal of Economic Literature* 52, no. 1 (2014): 5–44.
- Maćkowiak, Bartosz, and Mirko Wiederholt. “Optimal sticky prices under rational inattention”. *American Economic Review* 99, no. 3 (2009): 769–803.
- Malmendier, Ulrike, and Stefan Nagel. “Learning from inflation experiences”. *The Quarterly Journal of Economics* 131, no. 1 (2016): 53–87.
- Mankiw, N. Gregory, and Ricardo Reis. “Sticky information versus sticky prices: a proposal to replace the New Keynesian Phillips Curve”. *The Quarterly Journal of Economics* 117, no. 4 (2002): 1295–1328.
- Mankiw, N. Gregory, Ricardo Reis, and Justin Wolfers. “Disagreement about inflation expectations”. *NBER Macroeconomics Annual* 18 (2003): 209–248.
- Oster, Emily. “Unobservable selection and coefficient stability: Theory and evidence”. *Journal of Business & Economic Statistics* 37, no. 2 (2019): 187–204. <https://doi.org/10.1080/07350015.2016.1227711>.
- Reis, Ricardo. “Inattentive consumers”. *Journal of Monetary Economics* 53, no. 8 (2006): 1761–1800.
- Sims, Christopher A. “Implications of rational inattention”. *Journal of Monetary Economics* 50, no. 3 (2003): 665–690.
- Stock, James H., and Motohiro Yogo. “Testing for weak instruments in linear IV regression”. *Identification and Inference for Econometric Models* (2005): 80–108.
- Xie, Yu, and Jingwei Hu. “An introduction to the China Family Panel Studies (CFPS)”. *Chinese Sociological Review* 47, no. 1 (2014): 3–29.

Xie, Yu, Jingwei Hu, and Chunni Zhang. “CFPS: A longitudinal survey for studying development in China”. *China Economic Journal* 15, no. 1 (2022): 78–96.